Highlights

Density and graph estimation with smoothing splines and conditional Gaussian graphical models

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- We propose a new semiparametric density estimation framework called **C**ombined **S**moothing **S**pline and **C**onditional **G**aussian **G**raphical (cSScGG) model. cSScGG combines smoothing splines and cGGM, and thus is less prone to violation of parametric assumptions and more computationally efficient than nonparametric methods.
- We develop a graph estimation method based on the estimated density function. In addition to existing cGGM algorithms, we propose a geometric inference approach for edge detection. To the best of our knowledge, this is the first work that incorporates the smoothing spline technique into Gaussian graphical models.
- We establish convergence properties for both the overall density estimation and the graph estimation for the cGGM part. While we use the same cGGM model as [1], we consider a random design matrix, as well as distinct smoothing parameters for the penalties on the precision matrix and the matrix that parameterizes the conditional relationship between the design matrix and the response matrix.
- We proposed a backfitting algorithm for optimizing the penalized cGGM likelihood and selecting smoothing parameters, based on an approximation to a Leave-One-Out Kullback-Leibler (KL) divergence.

Density and graph estimation with smoothing splines and conditional Gaussian graphical models

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Abstract

Density estimation and graphical models play important roles in statistical learning. The estimated density can be used to construct a graphical model that reveals conditional relationships, whereas a graphical structure can be used to build models for density estimation. We propose a semiparametric framework that models part of the density function nonparametrically using a smoothing spline ANOVA (SS ANOVA) model and the conditional density parametrically using a conditional Gaussian graphical model (cGGM). This flexible framework allows us to deal with high-dimensional data without the Gaussian assumption. We develop computationally efficient algorithms for estimation and provide theoretical guarantees for our procedure. Our experimental results show that the proposed framework outperforms both parametric and nonparametric baselines.

Keywords: Density estimation; Graph estimation; Edge detection; High dimensional data; Penalized likelihood; Reproducing kernel Hilbert space; Smoothing spline ANOVA; Conditional Gaussian graphical Models

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1. Introduction

Density estimation has long been a subject of paramount interest in statistics and machine learning. Many parametric, nonparametric, and semiparametric methods have been developed. Semiparametric methods are a class of flexible models that take advantage of the parsimony of parametric models and the flexibility of nonparametric modeling. However, most existing semiparametric density estimation methods only focus on density estimation and fail to reveal the joint conditional independence network, unlike some parametric models such as the Gaussian graphical model (GGM) which can achieve both simultaneously. We aim to develop a new semiparametric procedure for density estimation that simultaneously explores the conditional independence structure in the model.

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We propose a semiparametric framework that models the joint density of a subset of the variables nonparametrically using a smoothing spline ANOVA (SS ANOVA) model and the conditional density of the rest of the variables parametrically using a conditional Gaussian graphical model (cGGM). The model we proposed is closely related to the semiparametric kernel density estimation (SKDE) ([2]) in which the same decomposition was considered. As stated in Hoti and Holmström [2], the crucial assumption in the proposed model is that the random vector can be split into non-Gaussian and conditionally Gaussian parts, and it is not realistic to expect such a split to be readily available in practice. Hoti and Holmström [2] proposed transforming data using PCA (Principal Component Analysis) or ICA (Independent Component Analysis). However, graph inference with PCA or ICA transformed data is not desirable and cumbersome to interpret. We propose to use the simple strategy of marginal test, which is to test each variable individually with normality tests such as the Shapiro-Wilks test to filter out the non-Gaussian variables. The idea here is not to find the accurate separation of variables, but to use nonparametric density and graph estimation where it matters most to improve the overall estimation.

The contribution of our paper lies in the following aspects:

1. We propose a new semiparametric density estimation framework called **C**ombined **S**moothing **S**pline and **C**onditional **G**aussian **G**raphical (cSScGG) model. cSScGG combines smoothing splines and cGGM, and thus is less prone to violation of parametric assumptions and more computationally efficient than nonparametric methods.
2. We develop a graph estimation method based on the estimated density function. On top of existing cGGM algorithms, we present a geometric inference approach to complete the edge detection. To the best of our knowledge, this is the first work that incorporates the smoothing spline technique into Gaussian graphical models.
3. We establish convergence properties for both the overall density estimation and the graph estimation for the cGGM part. While we use the same cGGM model as [1], we consider a random design matrix, as well as distinct smoothing parameters for the penalties on the precision matrix and the matrix that parameterizes the conditional relationship between the design matrix and the response matrix.
4. We proposed a backfitting algorithm for optimizing the penalized cGGM likelihood and selecting smoothing parameters, based on an approximation to a Leave-One-Out Kullback-Leibler (KL) divergence.

In Section 2, we introduce the semiparametric density model and methods for estimation and computation. We propose methods for graph estimation in Section 3. Section 4 presents theoretical results. Sections 5 and 6 show our simulation studies and applications to real datasets. In Section 7, we summarize our method. Technical details are in the Appendix.

2. Density Estimation with SS ANOVA and cGGM

2.1. Semiparametric Density Models with SS ANOVA and cGGM

Consider the density estimation problem in which we are given a random sample of a random vector $\boldsymbol{Z}$, and we wish to estimate the density function $f(\boldsymbol{z})$ of $\boldsymbol{Z}$. Let $\boldsymbol{Z} = (\boldsymbol{X}^T, \boldsymbol{Y}^T)^T$ where $\boldsymbol{X} = (X_1, \cdots, X_d)^T$ is a d-dimensional random vector for which the density function will be modeled nonparametrically and $\boldsymbol{Y} = (Y_1, \cdots, Y_p)^T \in \mathbb{R}^p$ collects elements for which the conditional density will be modeled parametrically. The joint density function $f(\boldsymbol{z})$ can be decomposed into two components:

$$f(\boldsymbol{z}) = f(\boldsymbol{x}, \boldsymbol{y}) = f(\boldsymbol{x})f(\boldsymbol{y}|\boldsymbol{x}). \tag{1}$$

We model $f(\boldsymbol{x})$ with SS ANOVA and $f(\boldsymbol{y}|\boldsymbol{x})$ with a cGGM, respectively.

Assume $\boldsymbol{X} \in \mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_d$ where $X_u \in \mathcal{X}_u \subset \mathbb{R}$. To relax the positivity and unity constraints of a density function, we consider the logistic transform $f = e^\eta / \int_{\mathcal{X}} e^\eta d\boldsymbol{x}$ where $\eta(\boldsymbol{x})$ is referred to as the logistic density function [3]. Assuming that η belongs to a tensor product of reproducing kernel Hilbert spaces (RKHS), the SS ANOVA decomposition of functions in the tensor product RKHS can be represented as

$$\eta(\boldsymbol{x}) = c + \sum_{k=1}^{d} \eta_k(x_k) + \sum_{k>l} \eta_{kl}(x_k, x_l) + \cdots + \eta_{1\ldots d}(x_1, \cdots, x_d), \tag{2}$$

where η_k's are main effects, η_{kl}'s are two-way interactions, and the rest are higher-order interactions involving more than two variables. Higher order interactions are often removed in Equation (2) for more tractable estimation and inference. An SS ANOVA model for the logistic density function assumes that η belongs to an RKHS which contains a subset of components in the above SS ANOVA decomposition. Terms included in the model can be regrouped, and the model space can be expressed as

$$\mathcal{H} = \mathcal{H}^0 \oplus \mathcal{H}^1 \oplus \cdots \oplus \mathcal{H}^w, \tag{3}$$

where $\mathcal{H}^0$ is a finite dimensional space, and $\mathcal{H}^1, \ldots, \mathcal{H}^w$ are orthogonal RKHS's with reproducing kernels (RK) R^v for $v = 1 \ldots, w$. Details about the SS ANOVA model can be found in Gu [3] and Wang [4].

For $f(\boldsymbol{y}|\boldsymbol{x})$, we assume that $\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x}$ follows $\mathrm{N}(-\Lambda^{-1}\Theta^T\boldsymbol{x}, \Lambda^{-1})$, where Λ is a $p\times p$ precision matrix and Θ is a $d\times p$ matrix that parameterizes the conditional relationship between $\boldsymbol{X}$ and $\boldsymbol{Y}$ [5, 1, 6]. Section 3.1 of the paper [5] detailed the motivation of the cGGM model. Essentially, the model is an equivalent reparmaterization of a multi-response regression model with $-\Lambda^{-1}\Theta^T$ as the matrix of regression coefficients and Λ^{-1} as the covariance matrix of the errors. The model is also motivated by the conditional Normal distribution $\boldsymbol{Y}$ given $\boldsymbol{X}$ when they are jointly Gaussian, which is, $\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x}$ follows $\mathrm{N}(\Sigma_{xy}^T\Sigma_{xx}^{-1}\boldsymbol{x}$, $\Sigma_{yy} - \Sigma_{xy}^T\Sigma_{xx}^{-1}\Sigma_{xy})$, where $\Sigma_{xy} = EX^TY$, $\Sigma_{xx} = EX^TX$, and $\Sigma_{yy} = EY^TY$. The model $\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x} \sim \mathrm{N}(-\Lambda^{-1}\Theta^T\boldsymbol{x}, \Lambda^{-1})$ is equivalent to the conditional Normal distribution when $\Lambda^{-1} = \Sigma_{yy} - \Sigma_{xy}^T\Sigma_{xx}^{-1}\Sigma_{xy}$ and $-\Lambda^{-1}\Theta^T = \Sigma_{xy}^T\Sigma_{xx}^{-1}$.

Now with the model for $f(\boldsymbol{x})$ and $f(\boldsymbol{y}|\boldsymbol{x})$, one can see from the density of $f(\boldsymbol{z})$, shown in (13), that Λ directly parameterizes the conditional independence among $\boldsymbol{Y}$, and Θ parameterizes the conditional independence between $\boldsymbol{X}$ and $\boldsymbol{Y}$ since when an element Θ_{ij} of Θ is zero, the likelihood is free of the interaction between x_i and y_j, implying that X_i and Y_j are conditionally independent. Note that the negative log likelihood is convex under this parameterization.

2.2. Penalized Likelihood Estimation

A cSScGG model has three parameters: $\eta \in \mathcal{H}$ and matrices Λ and Θ where $\mathcal{H}$ is an RKHS given in (3) and Λ is positive definite. Given an iid sample $\boldsymbol{Z}_i = (\boldsymbol{X}_i{}^T, \boldsymbol{Y}_i{}^T)^T$, $i = 1, \ldots, n$, let $X = (\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)^T$, $Y = (\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_n)^T$, $S_{xx} = n^{-1}X^TX$, $S_{yy} = n^{-1}Y^TY$, and $S_{xy} = n^{-1}X^TY$. Denote

$$\begin{aligned} l_1(\eta) &= \frac{1}{n}\sum_{i=1}^{n} e^{-\eta(\boldsymbol{X}_i)} + \int_{\mathcal{X}} \eta(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x}, & (4) \\ l_2(\Theta, \Lambda) &= -\log|\Lambda| + \operatorname{tr}(S_{yy}\Lambda + 2S_{xy}^T\Theta + \Lambda^{-1}\Theta^T S_{xx}\Theta) & (5) \end{aligned}$$

as the negative log pseudo-likelihood and negative log likelihood functions based on $\boldsymbol{X}$ and $\boldsymbol{Y}$ samples respectively, where some constants are ignored and ρ is a known density for the pseudo likelihood [7]. The pseudo-likelihood is a penalized M-estimation type method with a novel loss function that approximates the true log density with strong theoretical guarantees. It only involves one-dimensional integrations as opposed to multiple dimensional integrations in the true log density, which makes it particularly suitable for high dimensional $\boldsymbol{X}$. The density ρ is often chosen as the product of the marginal densities of the individual components in $\boldsymbol{X}$ [3], estimated from the data, which is what we used for the rest of this paper. The function $l_1(\eta)$ is continuous, convex and Fréchet differentiable, and $l_2(\Theta, \Lambda)$ is jointly convex in Λ and Θ.

We estimate η, Λ and Θ as minimizers of the penalized likelihood:

$$\{\hat{\eta}, \hat{\Lambda}, \hat{\Theta}\} = \underset{\eta\in\mathcal{H},\Lambda\succ 0,\Theta}{\arg\min} \left\{ \left[l_1(\eta) + \frac{\lambda_1}{2}J(\eta) \right] + \left[l_2(\Lambda, \Theta) + \lambda_2 \|\Lambda\|_{1,\text{off}} + \lambda_3 \|\Theta\|_1 \right] \right\}, \tag{6}$$

where J is a semi-norm in $\mathcal{H}$ that penalizes departure from the null space $\mathcal{H}^0$, $\|\cdot\|_1$ denotes the elementwise ℓ_1-norm, $\|\cdot\|_{1,\text{off}}$ denotes the elementwise ℓ_1-norm on off-diagonal entries, and $\Lambda \succ 0$ indicates positive definiteness of Λ. Together, $\|\Lambda\|_{1,\text{off}}$ and $\|\Theta\|_1$ encourage sparsity for the cGGM. We allow different tuning parameters for different penalties.

Note that the first part of the penalized likelihood depends on η only and the second part depends on Θ and Λ. Therefore, we can compute the penalized likelihood estimates by solving two optimization problems:

$$\hat{\eta} = \underset{\eta\in\mathcal{H}}{\arg\min} \left\{ \frac{1}{n}\sum_{i=1}^{n} e^{-\eta(\boldsymbol{X}_i)} + \int_{\mathcal{X}} \eta(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x} + \frac{\lambda_1}{2}J(\eta) \right\}, \tag{7}$$

and $\{\hat{\Theta}, \hat{\Lambda}\}$ as $\arg\min_{\Lambda\succ 0,\Theta} h(\Lambda, \Theta)$ where

$$h(\Lambda, \Theta) = \left\{ -\log|\Lambda| + \operatorname{tr}(S_{yy}\Lambda + 2S_{xy}^T\Theta + \Lambda^{-1}\Theta^T S_{xx}{}^T\Theta) + \lambda_2 \|\Lambda\|_{1,\text{off}} + \lambda_3 \|\Theta\|_1 \right\}. \tag{8}$$

As in Gu [3], we approximate the solution of (7) by a linear combination of basis functions in $\mathcal{H}^0$ and a random subset of representers. Then the estimate $\hat{\eta}$

can be calculated using the Newton-Raphson algorithm. The smoothing parameter λ_1 is selected as the minimizer of an approximated cross-validation estimate of the Kullback-Leibler (KL) divergence. Details are in Gu [3], Gu et al. [8], and Luo [9]. Next we propose a new computational method for optimizing (8).

2.3. Backfitting Algorithm for cGGM

In (8), when Θ and Λ are simultaneously updated as in Sohn and Kim [5], Wytock and Kolter [1] and Yuan and Zhang [6], an expensive computation of the large Hessian matrix of size $(p+d) \times (p+d)$ is required. In the following, we propose a backfitting procedure to update them iteratively, along with a few approximations to speed up the optimization of (8).

First, we present the backfitting algorithm in Algorithm 1.

Algorithm 1 backfitting algorithm for optimizing (8)

Require: S_{xx}, S_{xy}, S_{yy}, λ_2 and λ_3; Initial Λ_0, Θ_0, convergence threshold ϵ
Ensure: The minimum value in (8)
  Initialize $\Lambda \leftarrow \Lambda_0; \Theta \leftarrow \Theta_0$
  repeat
    Step A: Update $\Lambda_{t+1} \leftarrow h(\Lambda, \Theta_t)$
    Step B: Update $\Theta_{t+1} \leftarrow h(\Lambda_{t+1}, \Theta)$
  until convergence criterion $\|h(\Lambda_{t+1}, \Theta_{t+1}) - h(\Lambda_t, \Theta_t)\| < \epsilon$ is satisfied
  return Λ, Θ

In Step B, Θ can be updated efficiently using the coordinate descent algorithm, because with Λ fixed at $\Lambda_{(t)}$, Equation (8) becomes a quadratic function plus an ℓ_1 penalty.

However, for Step A, this is not the case. In the following, we propose an approximation for updating Λ in Step A. First, we consider the data as from an approximate conditional distribution $\mathrm{N}(-\Lambda_{(t)}^{-1}\Theta^T{}_{(t)}\boldsymbol{x}, \Lambda^{-1})$, whose negative log likelihood is proportional to

$$h_{(t)}(\Lambda) = -\log|\Lambda| + \operatorname{tr}\big(S_{yy}\Lambda + 2S_{xy}^T\Theta_{(t)}\Lambda_{(t)}^{-1}\Lambda + \Lambda_{(t)}^{-1}\Lambda\Lambda_{(t)}^{-1}\Theta^T{}_{(t)}S_{xx}\Theta_{(t)}\big), \quad (9)$$

Second, as in Hsieh et al. [10], we propose a quadratic approximation of $h_{(t)}$, based on the second-order Taylor expansion of $h_{(t)}(\Lambda)$ at $\Lambda_{(t)}$ where $\Lambda = \Lambda_{(t)} + \Delta_\Lambda$. Ignoring terms independent of Δ_Λ, the quadratic approximation is

$$\bar{h}_{(t)}(\Delta_\Lambda) = \operatorname{vec}(\nabla h_{(t)}(\Lambda_{(t)}))^T\operatorname{vec}(\Delta_\Lambda) + \frac{1}{2}\operatorname{vec}(\Delta_\Lambda)^T\nabla^2 h_{(t)}(\Lambda_{(t)})\operatorname{vec}(\Delta_\Lambda),$$

where $\nabla h_{(t)}(\Lambda_{(t)}) = S_{yy}+\Sigma_{(t)}\Theta_{(t)}^T S_{xx}\Theta_{(t)}\Sigma_{(t)}+2\Sigma_{(t)}\Theta_{(t)}^T S_{xy}-\Sigma_{(t)}$ and $\nabla^2 h_{(t)}(\Lambda_{(t)}) = \Sigma_{(t)} \otimes \Sigma_{(t)}$ are the gradient and Hessian matrices with respect to Λ, respectively, and $\otimes$ represents the Kronecker product.

Finally, we find Δ_Λ via the following and the coordinate descent algorithm [10]:

$$D_{\Lambda,(t)} = \arg\min_{\Delta_\Lambda}\left\{\bar{h}_{(t)}(\Delta_\Lambda) + \lambda_2\left\|\Lambda_{(t)} + \Delta_\Lambda\right\|_{1,\text{off}}\right\}. \quad (10)$$

Step A then is completed by the update: $\Lambda_{t+1} = \Lambda_t + \Delta_\Lambda$.

The regularized Newton step (10) is the most computational expansive part of the algorithm. Despite its efficiency for lasso type of problems, updating all

$p(p+1)/2$ variables in Λ is costly. To overcome this problem, we here propose our second approximation. We divide the parameter set into an active set and a free set. As in Hsieh et al. [10] and Wytock and Kolter [1], at the t-th iteration of the algorithm, we only update Θ and Λ over the active set defined by

$$\begin{aligned} \mathcal{S}_\Theta &= \{(i,j): \ |\left(\nabla_\Theta l_2(\Lambda_{(t)}, \Theta_{(t)})\right)_{ij}| > \lambda_3 \text{ or } \Theta_{ij,(t)} \neq 0\}, \\ \mathcal{S}_\Lambda &= \{(i,j): \ |\left(\nabla h_{(t)}(\Lambda_{(t)})\right)_{ij}| > \lambda_2 \text{ or } \Lambda_{ij,(t)} \neq 0\}. \end{aligned} \tag{11}$$

As the active set is relatively small due to sparsity induced by the ℓ_1 regularization, this strategy provides a substantial speedup.

Our third approximation is with regard to the tuning parameters λ_2 and λ_3. As a general selection tool, leave-one-out or k-fold cross-validation are usually used to select these tuning parameters, but it is computationally intensive. Various approximations have been proposed in the literature to mitigate this issue, including Lian [11] and Vujačić et al. [12]. The BIC and k-fold CV have been used to select a single tuning parameter in the cGGM [13, 5, 1, 6, 14]. The leave-one-out cross-validation (LOOCV) has not been used for the cGGM as it requires fitting the model n times. To the best of our knowledge, there is no computationally efficient alternative to LOOCV in the current cGGM literature.

To this end, we propose a new criterion, Leave-One-Out KL (LOOKL), as an approximation to LOOCV, which needs to fit the model only once. The derivation, shown in Luo [9], follows those in [11] and [12], by constructing an approximation to the LOOCV of the log-likelihood via Taylor expansions.

Note that the GACV in [11] and KLCV in [12] are LOOKL with $\Theta = 0$. The LOOKL selection criterion minimizes the following

$$\begin{aligned} &\text{LOOKL}(\lambda_2, \lambda_3) = -\tfrac{1}{n} l_2(\hat{\Lambda}, \hat{\Theta}) + \tfrac{1}{2n} \textstyle\sum_{k=1}^n \Big\{ \boldsymbol{u}_k^T (-C + B^T A^{-1} B)^{-1} \big[(-E + \\ &B^T A^{-1} D) \boldsymbol{v}_{xx,k} + 2 B^T A^{-1} \boldsymbol{v}_{xy,k} - \boldsymbol{v}_{yy,k}\big] + \ \boldsymbol{w}_k^T (-A + B C^{-1} B^T)^{-1} \\ &\big[(D - B C^{-1} E) \boldsymbol{v}_{xx,k} + B C^{-1} \boldsymbol{v}_{yy,k} - 2 \boldsymbol{v}_{xy,k}\big] \Big\}, \end{aligned} \tag{12}$$

where $S_{xx,k} = \boldsymbol{X}_k^T \boldsymbol{X}_k$, $S_{yy,k} = \boldsymbol{Y}_k^T \boldsymbol{Y}_k$, $S_{xy,k} = \boldsymbol{Y}_k^T \boldsymbol{X}_k$, $S_{xx}^{(-k)} = 1/n \sum_{i \neq k} S_{xx,i}$, $S_{yy}^{(-k)} = 1/n \sum_{i \neq k} S_{yy,i}$, $S_{xy}^{(-k)} = 1/n \sum_{i \neq k} S_{xy,i}$, $\boldsymbol{u}_k = \text{vec}(\hat{\Lambda}^{-1} - S_{yy,k} + \hat{\Lambda}^{-1} \hat{\Theta}^T S_{xx,k} \hat{\Theta} \hat{\Lambda}^{-1})$, $\boldsymbol{v}_{xx,k} = \text{vec}(S_{xx}^{(-k)} - S_{xx})$, $\boldsymbol{v}_{yy,k} = \text{vec}(S_{yy}^{(-k)} - S_{yy})$, $\boldsymbol{v}_{xy,k} = \text{vec}(S_{xy}^{(-k)} - S_{xy})$, $\boldsymbol{w}_k = \text{vec}(-2 S_{xy,k} - 2 S_{xx,k} \hat{\Theta} \hat{\Lambda}^{-1})$, $A = -2 \hat{\Lambda}^{-1} \otimes S_{xx}$, $B = 2 \hat{\Lambda}^{-1} \otimes S_{xx} \hat{\Theta} \hat{\Lambda}^{-1}$, $C = -\hat{\Lambda}^{-1} \otimes (\hat{\Lambda}^{-1} + 2 \hat{\Lambda}^{-1} \hat{\Theta}^T S_{xx} \hat{\Theta} \hat{\Lambda}^{-1})$, $D = -2 \hat{\Lambda}^{-1} \hat{\Theta}^T \otimes I_{d \times d}$, and $E = \hat{\Lambda}^{-1} \hat{\Theta}^T \otimes \hat{\Lambda}^{-1} \hat{\Theta}^T$.

Lastly, to initialize the backfitting algorithm, we propose the following. When $n > \max(p, d)$ and S_{xx} is invertible, we use the maximum likelihood estimates $\check{\Lambda} = (S_{yy} - S_{xy}^T S_{xx}^{-1} S_{xy})^{-1}$ and $\check{\Theta} = -S_{xx}^{-1} S_{xy} \check{\Lambda}$ of Λ and Θ as their initial values. Otherwise, the identity and zero matrix are used respectively [13].

3. Graph Estimation with cSScGG Models

Naturally, the edges among $\boldsymbol{Y}$ are identified by $\hat{\Lambda}$, and edges between $\boldsymbol{X}$ and $\boldsymbol{Y}$ are identified by $\hat{\Theta}$. We present the identification process of conditional independence within $\boldsymbol{X}$. Recall that the logarithm of the joint density is

$$\log f(\boldsymbol{z}) = \log f(\boldsymbol{x}) + \log f(\boldsymbol{y}|\boldsymbol{x}) \tag{13}$$
$$= \eta(\boldsymbol{x}) + \frac{1}{2}\big(- \boldsymbol{y}^T\Lambda\boldsymbol{y} - 2\boldsymbol{x}^T\Theta\boldsymbol{y} - \boldsymbol{x}^T\Theta\Lambda^{-1}\Theta^T\boldsymbol{x}\big) + C,$$

where C is a constant independent of $\boldsymbol{x}$ and $\boldsymbol{y}$. Let

$$\hat{\zeta}(\boldsymbol{x}) = \hat{\Delta}(\boldsymbol{x}) + \hat{\eta}(\boldsymbol{x}),$$

where $\hat{\Delta}(\boldsymbol{x}) = -\boldsymbol{x}^T\hat{\Theta}\hat{\Lambda}^{-1}\hat{\Theta}^T\boldsymbol{x}/2$. In the following we present the edge detection algorithm among $\boldsymbol{X}$ variables based on $\hat{\zeta}(\boldsymbol{x})$.

Gu [3] proposed a projection ratio to assess the importance of interaction terms given a non-parametric density. Specifically, for any pair of variables X and X', consider the function space with the decomposition: $\mathcal{S}^0 \oplus \mathcal{S}^1$, where $\mathcal{S}^0$ is the subspace of functions with main effects of X and X' only. .

Define the functional

$$\tilde{V}(f, g) = \textstyle\int_{\mathcal{X}} f(\boldsymbol{x})g(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x} - \{\int_{\mathcal{X}} f(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x}\}\{\int_{\mathcal{X}} g(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x}\} \tag{14}$$

and denote $\tilde{V}(f, f)$ as $\tilde{V}(f)$. $\tilde{V}(f - g)$ can be considered as a proxy of symmetrized KL divergence [3].

The squared error projection of any function $\zeta(X, X') \in \mathcal{S}^0 \oplus \mathcal{S}^1$ onto $\mathcal{S}^0$ is defined as[3]

$$\zeta_p = \underset{f\in\mathcal{S}^0}{\arg\min}\,\big\{\tilde{V}(\zeta - f)\big\}. \tag{15}$$

For any $\zeta_u \in \mathcal{S}^0$ such as a constant function, it is easy to check that $\tilde{V}(\zeta - \zeta_u) = \tilde{V}(\zeta - \zeta_p) + \tilde{V}(\tilde{\zeta} - \zeta_u)$. Then the projection ratio

$$r_{\mathcal{S}^0}(\zeta) = \tilde{V}(\zeta - \zeta_p)/\tilde{V}(\zeta - \zeta_u) \tag{16}$$

reflects the amount of contribution to ζ from the functions in $\mathcal{S}^1$, that is, the strength of the interaction between X and X'. A smaller ratio represents weaker strength. In our implementations, the threshold is set at 3%, that is, a projection ratio above the threshold indicates the existence of an edge. Details for computing the projection ratio are given in Luo [9].

We first rank all the pairs of the variables in $\boldsymbol{X}$ by their strength of interactions, that is, the projection ratio $r_{\mathcal{S}^0_{ij}}(\hat{\zeta})$ where $\mathcal{S}^0_{ij}$ is $\mathcal{H}$ excluding the interaction between X_i and X_j. Now we are ready to present our algorithm for identifying edges among $\boldsymbol{X}$ based on $\hat{\zeta}(\boldsymbol{x})$ as shown in Algorithm 2.

Finally to summarize, the conditional independence among $\boldsymbol{Y}$, between $\boldsymbol{X}$ and $\boldsymbol{Y}$, and among $\boldsymbol{X}$ are identified by zero elements in $\hat{\Lambda}$, $\hat{\Theta}$ and $\Pi = \big\{I\big(\mathcal{S}^0_s \text{ contains interaction between } X_i \text{ and } X_j\big), i, j = 1, ..., d\big\}$, respectively. The edge identification procedure is illustrated in Figure 1.

4. Theoretical Analysis

4.1. Notations and Assumptions

For a matrix U, let $|||U|||_2 = \sqrt{\lambda_{\max}(U^TU)}$, $|||U|||_\infty = \max_{i=1,\dots,p}\sum_{j=1}^p |U_{ij}|$ and $|||U|||_F = \sqrt{\sum_{i=1}^p\sum_{j=1}^p U_{ij}^2}$ be the ℓ_2 operator norm, ℓ_∞ operator norm and Frobenius norm respectively, where $\lambda_{\max}(U^TU)$ is the largest eigenvalue of U^TU. Let $S_{xx} = \frac{1}{n}X^TX, \Sigma = \Sigma_{xx} = E(\boldsymbol{X}\boldsymbol{X}^T)$, and $\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x} \sim \mathrm{N}(-\Lambda_0^{-1}\Theta_0^T\boldsymbol{x}, \Lambda_0^{-1})$ where Λ_0 and Θ_0 are the true parameters. Let $\Gamma_0 = (\Lambda_0^T, \Theta_0^T)^T$, $\Sigma_0 = \Lambda_0^{-1}$,

Algorithm 2 Edge identification among $\boldsymbol{X}$

Require: $\hat{\zeta}(\boldsymbol{x})$, ζ_u a constant function, projection ratio threshold c

Initialize $\mathcal{H} = \mathcal{S}^0 \oplus \mathcal{S}^1$ where $\mathcal{S}^0$ is the subspace spanned by all main effects of $\boldsymbol{X}$, and an overall projection ratio $r_{\mathcal{S}_0^0}(\hat{\zeta})$ with $\mathcal{S}_0^0 = \mathcal{S}^0$
if $r_{\mathcal{S}_0^0}(\hat{\zeta}) > c$ **then**
  repeat
    Step A: Identify the pairs whose interaction is not already in $\mathcal{S}_t^0$ and among these, choose the pair (k, l) with the largest ratio $r_{\mathcal{S}_{kl}^0}(\hat{\zeta})$
    Step B: update $\mathcal{S}_t^0$ with the addition of the interaction between X_k and X_l as $\mathcal{S}_{t+1}^0$
    Step C: calculate the overall projection ratio $r_{\mathcal{S}_{t+1}^0}(\hat{\zeta})$
  until convergence criterion $r_{\mathcal{S}_{t+1}^0}(\hat{\zeta}) < c$ is satisfied
end if
return $\mathcal{S}_{t+1}^0$ at convergence, denoted as $\mathcal{S}_s^0$.

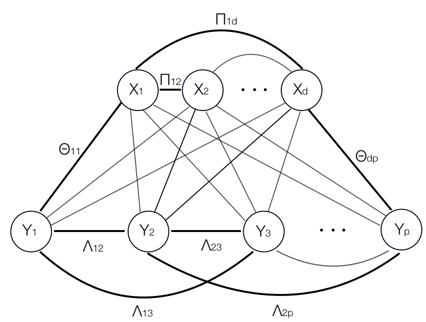

Figure 1: Illustration of the edge identification procedure.

$C_\sigma = \max_i \Sigma_{0,ii}$, $C_{\Sigma_0} = \max_{i,j} |\Sigma_{0,ij}|$, $C_\Theta = \max_{i,j} |\Theta_{0,ij}|$, $H_0 = \nabla_\Gamma^2 l_2(\Gamma_0)$ be the Hessian matrix evaluated at the true parameters, $H_{0,\Sigma} = E\nabla_\Gamma^2 l_2(\Gamma_0)$ and $\kappa_H = \max_{i,j} |H_{0,ij}^{-1}|$. Let $\gamma = \max_{1\le j\le p} \left\{ \sum_{i=1}^{d+p} I(\Gamma_{0,ij} \neq 0) \right\}$ be the maximum number of non-zeros in any column of Γ_0 representing the maximum degree of $\boldsymbol{Y}$ in the graph.

Let $\lambda = \max\{\lambda_2, \lambda_3\}$ and $r = \min\{\lambda_2, \lambda_3\}/\lambda$. We assume that there exists some constant C_0 such that $0 < C_0 \le r \le 1$. In the following, we assume that $\lambda_2 \ge \lambda_3$ and $r = \lambda_3/\lambda_2$. Similar arguments apply to the case of $\lambda_2 < \lambda_3$. We rewrite the objective function as

$$\{\hat{\Theta}, \hat{\Lambda}\} = \underset{\Lambda \succ 0, \Theta}{\arg\min} \left\{ l_2(\Lambda, \Theta) + \lambda(\|\Lambda\|_1 + r\,\|\Theta\|_1) \right\}. \tag{17}$$

We make the following assumptions.

Assumption 1. (Underlying Model) $\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x} \sim \mathrm{N}(-\Lambda_0^{-1}\Theta_0^T\boldsymbol{x}, \Lambda_0^{-1})$ where $\boldsymbol{Y}$ has the maximum degree γ.

Assumption 2. (Mutual incoherence) Let S denote the support set of Γ_0 in vector form $S = (vec(supp\{\Lambda_0\})^T, vec(supp\{\Theta_0\})^T)^T$ where $supp\{\cdot\}$ denotes the indicator function of whether an element is zero. Let $\bar{S}$ be the complement of S. Assume that $\left|\left|\left|\big(H_{0,\Sigma}\big)_{\bar{S}S}\big(H_{0,\Sigma})\big)^{-1}_{SS}\right|\right|\right|_\infty \le C_0(1-\alpha)$ for some $\alpha \in (0,1)$, where $\big(H_{0,\Sigma}\big)_{\bar{S}S}$ and $\big(H_{0,\Sigma}\big)_{SS}$ represent the $|\bar{S}| \times |S|$ and $|S| \times |S|$ sub-matrices of $H_{0,\Sigma}$ and having entries in $\bar{S} \times S$ and $S \times S$ respectively.

Assumption 3. (Control of eigenvalues) There exists some constants $0 < C_L \le C_U < \infty$, such that $C_L \le \lambda_{\min}(\Lambda_0) \le \lambda_{\max}(\Lambda_0) \le C_U$.

Assumption 1 provides the true underlying model. Assumption 2 limits the influence of edges in inactive set ($\bar{S}$) can have on the edges in active set (S), and Assumption 3 bounds the eigenvalues of the precision matrix. Define $V(f,g) = \int_{\mathcal{X}} f(\boldsymbol{x})g(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x}$ and $V(f) = \int_{\mathcal{X}} f^2(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x}$.

Assumption 4. V is completely continuous with respect to J and $J(\eta_0) < \infty$.

Under the Assumption 4, there exists ϕ_ν such that $V(\phi_\nu, \phi_\mu) = \delta_{\nu,\mu}$, $J(\phi_\nu, \phi_\mu) = \rho_\nu\delta_{\nu,\mu}$, and $0 \le \rho_\nu \uparrow \infty$, where $\delta_{\nu,\mu}$ is the Kronecker delta and ρ_ν are referred to as the eigenvalues of J with respect to V. Denote the Fourier series expansion of η_0 as $\eta_0 = \sum_\nu \eta_{\nu,0}\phi_\nu$ where $\eta_{\nu,0} = V(\eta_0, \phi_\nu)$ are the Fourier coefficients. Let $\tilde{\eta} = \sum_\nu \tilde{\eta}_\nu\phi_\nu$ where $\tilde{\eta}_\nu = (\beta_\nu + \eta_{\nu,0})/(1+\lambda_1\rho_\nu)$ and $\beta_\nu = n^{-1}\sum_{i=1}^n\{e^{-\eta_0(\boldsymbol{X}_i)}\phi_\nu(\boldsymbol{X}_i) - \int_{\mathcal{X}}\phi_\nu(\boldsymbol{x})\rho(\boldsymbol{x})d\boldsymbol{x}\}$.

Assumption 5. (a) The eigenvalues ρ_ν of J with respect to V satisfy $\rho_\nu > \beta\nu^s$ for some $\beta > 0$ and $s > 1$ when ν is sufficiently large.
(b) There exists some constants $0 < C_{1,1} < C_{1,2} < \infty$, $C_{1,3} < \infty$ and $C_{1,4} < \infty$ such that $C_{1,1} < e^{\eta_0(\boldsymbol{x})-\eta(\boldsymbol{x})} < C_{1,2}$ holds uniformly for η in a convex set around η_0 containing $\hat{\eta}$ and $\tilde{\eta}$, $e^{-\eta_0(\boldsymbol{x})} < C_{1,3}$, and $\int_{\mathcal{X}}\phi_\nu^2(\boldsymbol{x})\phi_\mu^2(\boldsymbol{x})e^{-\eta_0(\boldsymbol{x})}\rho(\boldsymbol{x})d\boldsymbol{x} < C_{1,4}$ for any ν and μ.
(c) There exists some $q \in [1,2]$ such that $\sum_\nu \rho_\nu^q\eta_{\nu,0}^2 < \infty$.

Assumptions 4 and 5 are commonly used in smoothing spline literature to study the convergence rate for nonparametric density estimation [3] .

4.2. Asymptotic Consistency of the Estimated Parameters

Before presenting the theorem for the estimation error bound and edge selection accuracy, we introduce two lemmas for our random design setting of the cGGM, where $\boldsymbol{X}$ is random.

Lemma 1. *Suppose that $||\boldsymbol{X}_i||_2 \le C_1\sqrt{d}$ a.s. and $E|\langle\boldsymbol{X}_i, \boldsymbol{x}\rangle|^q \le C_2^q$ for any $\boldsymbol{x} \in \Omega$, $i = 1, \cdots, n$, $d < n$, $q > 4$, and some constants C_1 and C_2, where $\boldsymbol{X}_i$ is the i-th row of X, and $\Omega = \{\boldsymbol{x} \in \mathcal{X} : ||\boldsymbol{x}||_2^2 = 1\}$. Let $\epsilon > 0$ and*

$$n_0 = \Big(\max\{C_*, C_{**}\}(\log\log d)^2 d^{1/2-2/q}\Big)^{-1/(2/q-1/2)},$$

where C_* is a constant depending on q, C_1, C_2, and ϵ only, $C_{**} = C_\alpha^{-1}C_HC_*$, $C_\alpha = (\alpha - \alpha(2-\alpha)/4)/(\alpha/4 + 1/C_0)$ for $\alpha \in (0,1)$, C_H is a constant related to

$H_{0,\Sigma}$ (see the proof of Lemma 1 in Appendix A). Then, for any $\epsilon > 0$, when $n > n_0$, with probability at least $1-\epsilon$, we have

$$|||\big(H_{0,\Sigma}\big)_{\bar{S}S}\big(H_{0,\Sigma}\big)_{SS}^{-1} - \big(H_0\big)_{\bar{S}S}\big(H_0\big)_{SS}^{-1}|||_\infty < C_\alpha.$$

Lemma 2. *Suppose that for the jth column $\boldsymbol{X}^j$, $\sigma_{4,j}^2 = Var[X_{ij}^2]$ is finite, for $i \in \{1,\cdots,n\}$. Denote $C_s^2 = \max_j\{\Sigma_{jj}\}$, where Σ_{jj} is the diagonal element of Σ, for $j = 1,\cdots,d$. Let $\epsilon > 0$, $C_X > C_s$ be a finite constant, and $n_c = \max_j\left\{\frac{\sigma_{4,j}^2}{\epsilon(C_X^2-\Sigma_{jj})^2}\right\}$. Then, we have $\max_{j=1,\ldots,d}\left\|\boldsymbol{X}^j\right\|_2/\sqrt{n} \le C_X$ with at least probability $1-\epsilon$ when the sample size $n > n_c$.*

The following theorem provides the estimation error bound and edge selection accuracy.

Theorem 1. *Suppose that the Assumption 2, and Lemmas 1 and 2 hold, $\tau > 2$, $d < n$, and*

$$\begin{aligned} n &\geq \max\left\{C_{2,1}C_{2,2}^2C_\sigma^2\gamma^4(1+8\alpha^{-1}C_0^{-1})^4[\tau\log(pd)+\log 4], n_0, n_c\right\}, \quad (18)\\ \lambda &= 8\alpha^{-1}C_0^{-1}C_\sigma C_s^\star\sqrt{3200}\sqrt{\frac{\tau\log(pd)+\log 4}{n}}, \end{aligned}$$

where $C_{2,1} = \max\{12800, 32C_s^2\}$, $C_{2,2} = \kappa_H\max\{3C_{\Sigma_0}/\gamma, 2/(C_\Theta\gamma), 412C_{\Sigma_0}^4C_\Theta^2C_X^2\}$, and $C_s^\star = \max\{C_s, 1\}$. Then we have

1. *With probability greater than $\Big(1-\big(p^{-(\tau-2)}+(pd)^{-(\tau-1)}\big)\Big)\Big(1-\epsilon\Big)$, the estimates satisfy the elementwise ℓ_∞ bound:*

$$\max\left\{\left\|\hat{\Lambda}-\Lambda_0\right\|_\infty, \left\|\hat{\Theta}-\Theta_0\right\|_\infty\right\} \leq$$
$$2\kappa_H(1+8\alpha^{-1}C_0^{-1})C_\sigma C_s^\star\sqrt{3200}\sqrt{\frac{\tau\log(pd)+\log 4}{n}}.$$

2. *With probability greater than $\Big(1-\big(p^{-(\tau-2)}+(pd)^{-(\tau-1)}\big)\Big)(1-\epsilon)^2$, all non-zero entries of the solution $(\hat{\Lambda},\hat{\Theta})$ are a subset of the non-zero entries of (Λ_0,Θ_0). Furthermore, non-zero entries of $(\hat{\Lambda},\hat{\Theta})$ includes all non-zero entries (i,j) in (Λ_0,Θ_0) that satisfy*

$$\min\{\Lambda_{0,ij},\Theta_{0,ij}\} > 4\kappa_H(1+8\alpha^{-1}C_0^{-1})C_\sigma C_s^\star\sqrt{3200}\sqrt{\frac{\tau\log(pd)+\log 4}{n}}, \quad (19)$$

where ϵ is a fixed accuracy close to zero.

Remark 1: i) Theorem 1 indicates that a sample size larger than n_0, n_c, and a constant times $\gamma^4\log(pd)$ is enough for our estimation procedure to identify a subset of the true non-zero elements in the cGGM, and the resulting estimations are close to the true parameters in ℓ_∞ bound. The additional n_0 and n_c are

due to the introduction of the random design X. Otherwise, the convergence rate is the same as that in Wytock and Kolter [1], but the success probability of the primal-dual witness approach as well as the exact bounds for n and λ are different. We also provide a lower bound for the sign consistency, which is not included in Wytock and Kolter [1]. Note in Wytock and Kolter [1], $C_0 = 1$ due to the use of a single penalty parameter, and the factor $(1+8\alpha^{-1}C_0^{-1})$ is on the order of 2 instead of 4 here, which we believe was a mistake in their paper.
ii) The convergence probability is smaller than that for the GGM [15] where only a precision matrix is estimated, whereas we have an extra parameter Θ.

Define s_Λ as the total number of non-zero elements in off-diagonal positions of Λ_0, and s_Θ as the total number of non-zero elements in Θ_0.

Corollary 1. *Under the same assumptions as in Theorem 1, with probability at least* $\Big(1 - \big(p^{-(\tau-2)} + (pd)^{-(\tau-1)}\big)\Big)(1-\epsilon)$, *the estimates* $\hat{\Lambda}$ *and* $\hat{\Theta}$ *satisfy*

$$\begin{aligned}&\max\left\{\left\|\left|\hat{\Lambda} - \Lambda_0\right|\right\|_F, \left\|\left|\hat{\Theta} - \Theta_0\right|\right\|_F\right\}\\ \le& 2\kappa_H(1+8\alpha^{-1}C_0^{-1})\max\{\sqrt{p+s_\Lambda}, \sqrt{s_\Theta}\}C_\sigma C_s^\star\sqrt{3200}\sqrt{\frac{\tau\log(pd)+\log 4}{n}}.\end{aligned} \tag{20}$$

Remark 2: The Frobenius norm was not studied in Wytock and Kolter [1]. It is useful for the convergence rate for the density estimation in Section 4.3.

4.3. Convergence Rates for the Density Estimation

We first introduce a combined measure of divergence between the joint density and its estimate. Let $f_0(\boldsymbol{x}) = e^{\eta_0(\boldsymbol{x})}\rho(\boldsymbol{x})/\int_{\mathcal{X}} e^{\eta_0(\boldsymbol{x})}\rho(\boldsymbol{x})d\boldsymbol{x}$ and $f_0(\boldsymbol{y}|\boldsymbol{x})$ be the true densities of $\boldsymbol{X}$ and $\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x}$ with their estimates denoted as $\hat{f}(\boldsymbol{x}) = e^{\hat{\eta}(\boldsymbol{x})}\rho(\boldsymbol{x})/\int_{\mathcal{X}} e^{\hat{\eta}(\boldsymbol{x})}\rho(\boldsymbol{x})d\boldsymbol{x}$ and $\hat{f}(\boldsymbol{y}|\boldsymbol{x})$ respectively. The KL divergence between two density functions f_1 and f_2 is defined as $\mathrm{KL}(f_1, f_2) = \mathrm{E}_{f_1}[\log(f_1/f_2)]$. Then the symmetrized KL divergence between the true joint density $f_0(\boldsymbol{z}) = f_0(\boldsymbol{x})f_0(\boldsymbol{y}|\boldsymbol{x})$ and its estimate $\hat{f}(\boldsymbol{z}) = \hat{f}(\boldsymbol{x})\hat{f}(\boldsymbol{y}|\boldsymbol{x})$ is

$$\begin{aligned}&\mathrm{SKL}\Big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\Big) = \mathrm{KL}\Big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\Big) + \mathrm{KL}\Big(\hat{f}(\boldsymbol{z}), f_0(\boldsymbol{z})\Big)\\ &= \left\{\int_{\mathcal{X}} f_0(\boldsymbol{x})\mathrm{KL}\Big(f_0(\boldsymbol{y}|\boldsymbol{X}=\boldsymbol{x}), \hat{f}(\boldsymbol{y}|\boldsymbol{X}=\boldsymbol{x})\Big)d\boldsymbol{x} + \int_{\mathcal{X}} \hat{f}(\boldsymbol{x})\mathrm{KL}\Big(\hat{f}(\boldsymbol{y}|\boldsymbol{X}=\boldsymbol{x}),\right.\\ &\left. f_0(\boldsymbol{y}|\boldsymbol{X}=\boldsymbol{x})\Big)d\boldsymbol{x}\right\} + \left\{\mathrm{KL}\Big(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x})\Big) + \mathrm{KL}\Big(\hat{f}(\boldsymbol{x}), f_0(\boldsymbol{x})\Big)\right\}\\ &\triangleq \mathrm{SKL}\big(f_0(\boldsymbol{y}|\boldsymbol{x}), \hat{f}(\boldsymbol{y}|\boldsymbol{x})\big) + \mathrm{SKL}\big(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x})\big).\end{aligned} \tag{21}$$

We will establish the asymptotic convergence rate under the following

$$D\big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\big) \triangleq \mathrm{SKL}\big(f_0(\boldsymbol{y}|\boldsymbol{x}), \hat{f}(\boldsymbol{y}|\boldsymbol{x})\big) + (V + \lambda_1 J)(\eta_0 - \hat{\eta}). \tag{22}$$

The difference between (22) and (21) lies in the divergence measures for the estimation of $f(\boldsymbol{x})$. Note that the rate in $V(\eta-\eta_0)$ implies rate in $\tilde{V}(\eta-\eta_0)$, since $\tilde{V}(f) \le V(f)$ and $\tilde{V}(\eta_0 - \hat{\eta})$ is a proxy of $\mathrm{SKL}(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x}))$ [3].

Theorem 2. *Under Assumption 3 and* $\left\|\left|\hat{\Lambda} - \Lambda_0\right|\right\|_F \leq 0.5 C_L$, *we have*

$$SKL\big(f_0(\boldsymbol{y}|\boldsymbol{x}), \hat{f}(\boldsymbol{y}|\boldsymbol{x})\big) = \mathcal{O}\Big(n^{-5/2} p^{5/2} (\log pd)^{5/2} + n^{-1} p^2 (\log pd)\Big). \tag{23}$$

For the smoothing spline ANOVA estimate $\hat{\eta}$ of η_0, under the Assumptions 4 and 5, Gu [3] showed that as $\lambda_1 \to 0$ and $n\lambda_1^{2/s} \to \infty$,

$$(V + \lambda_1 J)(\hat{\eta} - \eta_0) = \mathcal{O}(n^{-1}\lambda_1^{-1/s} + \lambda_1^q). \tag{24}$$

Finally, we have the convergence rate for the joint density estimate.

Theorem 3. *Suppose that the Assumptions 2-5, Lemmas 1 and 2 hold,* $d < n$ $\tau > 2$, $\lambda_1 \to 0$, $n\lambda_1^{2/s} \to \infty$, *and*

$$\begin{aligned} n &\geq C_{3,1} C_{3,2}^2 C_{\Sigma_0}^2 \max\{\gamma^4, p + s_\Lambda\}(1 + 8\alpha^{-1} C_0^{-1})^4 [\tau \log(pd) + \log 4], \quad (25) \\ \lambda &= 8\alpha^{-1} C_\sigma C_s^\star \sqrt{3200} \sqrt{\frac{\tau \log(pd) + \log 4}{n}}, \end{aligned}$$

where $C_{3,1} = C_{2,1} = \max\{12800, 32C_s^2\}$, *and* $C_{3,2} = \max\{C_{2,2}, \kappa_H \sqrt{1600}/C_L\} = \kappa_H \max\{\, 3C_{\Sigma_0}/\gamma, 2/(C_\Theta \gamma), 412 C_{\Sigma_0}^4 C_\Theta^2 C_X^2, \sqrt{1600}/C_L\}$, *then with probability greater than* $\Big(1 - \big(p^{-(\tau-2)} + (pd)^{-(\tau-1)}\big)\Big)(1 - \epsilon)$, *we have*

$$D\big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\big) = \mathcal{O}\big(n^{-5/2} p^{5/2} (\log pd)^{5/2} + n^{-1} p^2 (\log pd) + n^{-1}\lambda_1^{-1/s} + \lambda_1^q\big).$$

Remark 3: In the case of low-dimensional $\boldsymbol{X}$ (usually $d \leq 3$), the computation of multivariate integrals are feasible. We may use the penalized likelihood instead of the pseudo likelihood to estimate the density function $f(\boldsymbol{x})$. This leads to $f_0(\boldsymbol{x}) = e^{\eta_0} / \int_{\mathcal{X}} e^{\eta_0}$. Under similar conditions, Gu [3] has proved that the symmetrized KL divergence $\mathrm{SKL}(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x}))$ is also $\mathcal{O}(n^{-1}\lambda_1^{-1/s} + \lambda_1^q)$, where $\hat{f}(\boldsymbol{x})$ is the penalized likelihood estimate. If we also use the penalized likelihood to estimate η in our model, then $\mathrm{SKL}\big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\big) = \mathcal{O}(n^{-5/2} p^{5/2} (\log pd)^{5/2} + n^{-1} p^2 (\log pd) + n^{-1}\lambda_1^{-1/s} + \lambda_1^q)$.

5. Simulation Studies

For density estimation, we use both LOOKL and CV (5-fold) methods to choose λ_2 and λ_3. Tuning parameters involved in all other methods are chosen by 5-fold CV. For graph estimation, we select λ_2 and λ_3 in the cSScGG method as minimizers of the BIC score:

$$\mathrm{BIC}(\lambda_2, \lambda_3) = \Big\{-n \log |\hat{\Lambda}| + n \mathrm{tr}(S_{yy}\hat{\Lambda} + 2S_{xy}^T \hat{\Theta} + \hat{\Lambda}^{-1} \hat{\Theta}^T S_{xx} \hat{\Theta})\Big\} + \log n \{\xi(\hat{\Lambda})/2 + \xi(\hat{\Theta})\}, \tag{26}$$

where $\xi(\hat{\Lambda})$ and $\xi(\hat{\Theta})$ are the number of non-zero off-diagonal elements in $\hat{\Lambda}$ and the number of non-zero elements in $\hat{\Theta}$. The degree of freedom is defined the same as in Yin and Li [13]. The BIC is also used to select tuning parameters in other methods for graph estimation. The R code for the simulations can be found at `https://github.com/annaliuumass/cSScGG`.

5.1. Density Estimation

We note that the semiparametric cSScGG method has similar KLs as the parametric QUIC method when $\mathbf{X}$ is Gaussian (not shown, see Luo [9]). In the following, we set $n = 200$, $d = 3$, $p = 20$, and generate $\boldsymbol{X} \sim \omega\mathrm{N}(\boldsymbol{\mu}_1, \sigma^2 I) + (1-\omega)\mathrm{N}(\boldsymbol{\mu}_2, \sigma^2 I)$ with $\boldsymbol{\mu}_1 = (1, 0, -1)^T$, and $\boldsymbol{\mu}_2 = (0, -1, 1)^T$. We consider four combinations of σ and ω: $\sigma = 0.5, 0.1$ and $\omega = 0.9, 0.5$. Results are based on 100 replications under each setting. In each replication, we first generate n iid samples $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ from the multivariate Gaussian mixtures, then $\boldsymbol{Y}_i$'s are generated from a cGGM. Specifically, we randomly create a $(d+p) \times (d+p)$ precision matrix Ω using the R-package `huge` [16], in which the probability of the off-diagonal elements being nonzero equals 0.2. The decomposition $\Omega = \begin{bmatrix} \Omega_{xx} & \Omega_{xy} \\ \Omega_{yx} & \Omega_{yy} \end{bmatrix}$ gives us $\Theta = \Omega_{xy}$ and $\Lambda = \Omega_{yy}$, so that we can sample $\boldsymbol{Y}_i$ from $\mathrm{N}(-\Lambda^{-1}\Theta^T\boldsymbol{X}_i, \Lambda^{-1})$ for $i = 1, \ldots, n$.

Since the division of non-Gaussian variables $\boldsymbol{X}$ and Gaussian variables $\boldsymbol{Y}$ is typically unknown in practice; we consider two versions of the proposed method – plain cSScGG and cSScGG with normality test (denoted as NT). In the plain version, we assume that the true non-Gaussian components are known and apply cSScGG directly. In the NT version, we select variables with p-values based on the Shapiro-Wilk test for all $p + d$ marginal variables that are less than $0.05/(p + d)$ as $\boldsymbol{X}$, and then apply the cSScGG method.

We also estimate density using the SKDE [2], MLE, and QUIC [10] methods. In the implementation of the SKDE method, we use the R-package `ks` [17] to calculate the kernel density estimate for $f(\boldsymbol{x})$ with the bandwidth selected by the smoothed cross-validation selector with diagonal bandwidth matrices (`Hscv.diag(x)`), which provides the best overall performance. To avoid selecting the two extra bandwidths involved in SKDE, as in Hoti and Holmström [2], we set $f(\boldsymbol{y}|\boldsymbol{x}) = f(\boldsymbol{y})$ and use MLE to estimate $f(\boldsymbol{y})$. MLE and QUIC methods treat $\boldsymbol{Z}^T = (\boldsymbol{X}^T, \boldsymbol{Y}^T)$ as multivariate Gaussian across all settings, and the estimates from these two methods are further broken down into $f(\boldsymbol{x})$ and $f(\boldsymbol{y}|\boldsymbol{x})$ for comparison. Specifically, QUIC method learns the precision matrix of $\boldsymbol{Z}$ by forming a quadratic approximation of the log-likelihood, and the estimates are computed using the R-package `QUIC` [18].

To evaluate the performance of different methods, we consider the KL divergence between the estimated density and the true density

$$\mathrm{KL}\Big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\Big) = \mathrm{E}_{\boldsymbol{X}}\Big[\mathrm{KL}\Big(f_0(\boldsymbol{y}|\boldsymbol{X}), \hat{f}(\boldsymbol{y}|\boldsymbol{X})\Big)\Big] + \mathrm{KL}\Big(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x})\Big),$$

where f_0 is the true density, and the aggregated KL, $\mathrm{E}_{\boldsymbol{X}}\Big[\mathrm{KL}\Big(f_0(\boldsymbol{y}|\boldsymbol{X}), \hat{f}(\boldsymbol{y}|\boldsymbol{X})\Big)\Big]$, is approximated by the empirical aggregated KL. Table 1 reports the overall KL divergence $\mathrm{KL}\Big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\Big)$, the empirical aggregated KL divergence $\frac{1}{n}\sum_{i=1}^{n} \mathrm{KL}\ \Big(f_0(\boldsymbol{y}|\boldsymbol{X}_i),\ \hat{f}(\boldsymbol{y}|\boldsymbol{X}_i)\Big)$, and $\mathrm{KL}\Big(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x})\Big)$. They provide evaluations for the estimation of $f(\boldsymbol{z})$, $f(\boldsymbol{y}|\boldsymbol{x})$, and $f(\boldsymbol{x})$, respectively. Since cSScGG with normality test may identify different $\boldsymbol{X}$, we only included it in the overall KL divergence $\mathrm{KL}\Big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\Big)$ comparison. Both versions of the cSScGG

Table 1: Averages and standard deviations (in parentheses) of the KL divergence. NT means that the Shapiro-Wilk normality test was applied for selecting $\boldsymbol{X}$ variables.

KL	Method	$\sigma = 0.5$ $\omega = 0.9$	$\sigma = 0.5$ $\omega = 0.5$	$\sigma = 0.1$ $\omega = 0.9$	$\sigma = 0.1$ $\omega = 0.5$
$\mathrm{KL}\Big(f_0(\boldsymbol{x}), \hat{f}(\boldsymbol{x})\Big)$ for $f(\boldsymbol{x})$	**cSScGG**	**0.03(0.02)**	**0.04(0.02)**	**0.06(0.05)**	**0.04(0.03)**
	SKDE	0.13 (0.02)	0.15 (0.02)	0.56 (0.35)	0.80 (0.03)
	QUIC	0.23 (0.02)	0.24 (0.01)	3.33 (0.04)	3.53 (0.02)
	MLE	0.18 (0.02)	0.22 (0.01)	3.14 (0.04)	3.48 (0.02)
$\frac{1}{n}\sum_{i=1}^{n} \mathrm{KL}\Big(f_0(\boldsymbol{y}\|\boldsymbol{X}_i), \hat{f}(\boldsymbol{y}\|\boldsymbol{X}_i)\Big)$ for $f(\boldsymbol{y}\|\boldsymbol{x})$	**cSScGG_CV**	**0.53(0.12)**	**0.59(0.19)**	**0.49(0.10)**	**0.55(0.15)**
	cSScGG_LOOKL	0.86 (0.12)	0.85 (0.14)	0.93 (0.13)	0.88 (0.13)
	SKDE	1.01 (0.17)	1.18 (0.22)	0.85 (0.13)	1.04 (0.18)
	QUIC	0.87 (0.13)	0.90 (0.15)	0.81 (0.11)	0.87 (0.14)
	MLE	1.00 (0.17)	1.07 (0.22)	0.94 (0.13)	1.01 (0.18)
$\mathrm{KL}\Big(f_0(\boldsymbol{z}), \hat{f}(\boldsymbol{z})\Big)$ for $f(\boldsymbol{z})$	cSScGG_CV	0.55 (0.12)	0.63 (0.19)	0.54 (0.12)	0.59 (0.15)
	cSScGG_CV_NT	**0.51(0.17)**	**0.53(0.09)**	**0.49(0.39)**	**0.54(0.09)**
	cSScGG_LOOKL	0.89 (0.12)	0.89 (0.14)	0.99 (0.14)	0.92 (0.13)
	cSScGG_LOOKL_NT	0.85 (0.15)	0.87 (0.10)	0.86 (0.40)	0.84 (0.11)
	SKDE	1.14 (0.17)	1.32 (0.22)	1.40 (0.38)	1.83 (0.19)
	QUIC	1.09 (0.13)	1.14 (0.15)	4.13 (0.13)	4.40 (0.14)
	MLE	1.19 (0.17)	1.30 (0.22)	4.08 (0.14)	4.50 (0.18)

method enjoy superior performance compared to all other methods in all settings. Interestingly, the NT version, where the division of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is unknown, shows a slightly better overall KL divergence on average. When comparing the plain cSScGG with other methods, the differences mainly come from the estimation of $f(\boldsymbol{x})$, in which parametric methods MLE and QUIC cannot fit the data properly. The cSScGG performs much better than SKDE in both the estimation of $f(\boldsymbol{x})$ and $f(\boldsymbol{y}|\boldsymbol{x})$. When σ is fixed, larger differences are observed under $\omega = 0.5$ with more severe deviation from Gaussian. Furthermore, for a fixed ω, the superiority of the cSScGG methods is greater when $\sigma = 0.1$ with more severe deviation from Gaussian.

5.2. Edge Detection

We do not consider the SKDE and MLE methods here because they do not perform edge selection. We compare with the QUIC method for the Gaussian graphical model, which provides similar results as the Glasso. We also include the nonparanormal (NPN) method [19], which is implemented with the R-package `huge`. When fitting the model, we use shrunken ECDF to transform the data first, then apply Glasso to the transformed data. The final NPN model is selected by the extended BIC score [20].

We consider two scenarios: 1) $\boldsymbol{X}$ and $\boldsymbol{Y}$ are from a joint multivariate Gaussian distribution, and 2) $\boldsymbol{X}$ is generated from a non-Gaussian distribution and $\boldsymbol{Y}$ is Gaussian conditional on $\boldsymbol{X}$. In both scenarios, our edge detection procedure is agnostic to what variables constitute $\boldsymbol{X}$ and select $\boldsymbol{X}$ variables using the same marginal approach, that is, a normality test such as Shapiro-Wilk test is applied to each variable and those with a p-value of less than $0.05/(p+d)$ are considered to belong to $\boldsymbol{X}$.

In the first scenario, we generate a $(d+p)\times(d+p)$ sparse precision matrix Ω, in which the probability of the off-diagonal elements being nonzero equals 0.2. Then n i.i.d. samples $\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_n$ are generated from $\mathrm{N}(\mathbf{0}, \Omega^{-1})$.

In the second scenario, we first generate samples from the density:

$$
\begin{aligned}
&f(x_1, x_2, x_3, c_1, c_2) \propto \exp(-c_2(x_3 - 0.8)^2) \\
&\big(\exp(-5(c_1 x_1 - 0.2x_3 - 0.2)^2) + 2\exp(-5(c_1 x_1 - 0.2x_3 - 0.6))^2\big) \\
&\big(\exp(-50(c_1 x_2 - 0.4x_3 - 0.2)^2) + 2\exp(-50(c_1 x_2 - 0.4x_3 - 0.6)^2)\big)\,,
\end{aligned}
\tag{27}
$$

where $0 < x_1, x_2, x_3 < 1$, and $c_1 = 0.5, c_2 = 10$ or $c_1 = 1, c_2 = 1$. Let $\boldsymbol{X}$ be the centered samples with zero expectations. $\boldsymbol{Y}$ is then generated as $\boldsymbol{Y}|\boldsymbol{X} \sim \mathrm{N}(-\Lambda^{-1}\Theta^T\boldsymbol{X}, \Lambda^{-1})$, with Λ and Θ obtained the same way as in the first scenario and in Section 5.1.

In both scenarios, we fix $p = 20$, $d = 3$, and consider two sample sizes $n = 200$ and $n = 300$. In the first scenario, when $\mathbf{Z}$ follows the joint Gaussian distribution, the normality test with a cutoff of $0.05/(p+d)$ doesn't select any non-Gaussian variables 95% of the time, and cSScGG is the same as QUIC when it happens. To meaningfully compare the methods in this scenario, we randomly chose three variables as $\mathbf{X}$ and fit cSScGG. In the second scenario with $c_1 = 0.5, c_2 = 10$, the normality test selects the three $\mathbf{X}$ variables with a specificity of 0.998 and a sensitivity of 0.67. In the second scenario with $c_1 = 1, c_2 = 1$, the normality test selects the three $\mathbf{X}$ variables with a specificity of 0.996 and a sensitivity of 1.

We compare our method against QUIC and NPN based on the performance on the overall edge detection of the whole graph of $\boldsymbol{Z}^T = (\boldsymbol{X}^T, \boldsymbol{Y}^T)$, in terms of the specificity (SPE), sensitivity (SEN) and F_1 score (Figure 2). For the non-Gaussian case (27) with $c_1 = 1$ and $c_2 = 1$, we also break down the comparison into between $\boldsymbol{X}$, between $\boldsymbol{Y}$, and between $\boldsymbol{X}$ and $\boldsymbol{Y}$ (Figure 3).

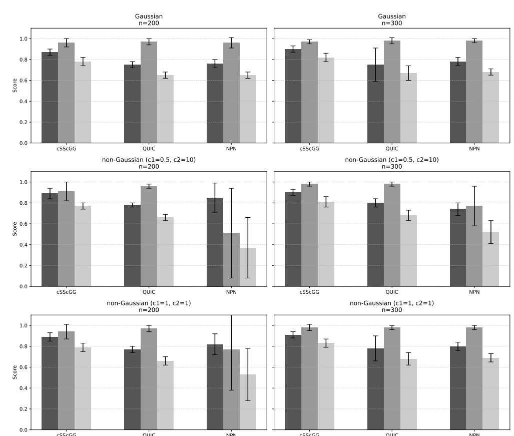

Figure 2: Performance metrics (SPE (dark), SEN (gray), F1(light)) across methods and sample sizes on the overall edge detection of the whole graph of $\boldsymbol{Z}$.

Overall, the cSScGG performs better than the QUIC and NPN method, even when the normality assumption holds or when the non-Gaussian variables were not accurately identified. It suggests that the flexibility and the semi-parametric nature of the cSScGG method, including the nonparametric density estimation and the projection ratio method for variables deviating from the Gaussian distribution, and the conditional Gaussian model for the Gaussian variables with two smoothing parameters, allows us to better model the conditional independence

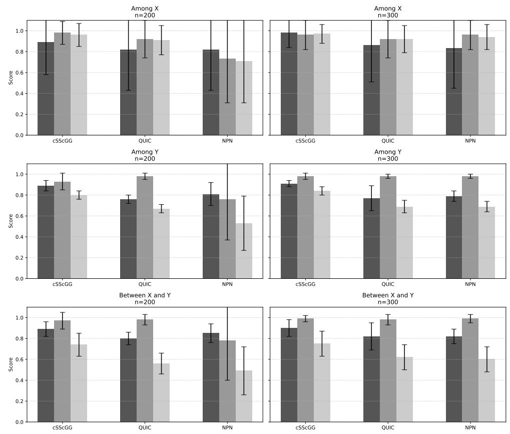

Figure 3: Comparison of SPE (dark), SEN (gray), and F1 (light) scores among and between X and Y for different methods for the non-Gaussian case (27) with $c_1 = 1$ and $c_2 = 1$.

among the variables.

5.3. Empirical rates of the data-driven smoothing parameters

Theorem 1 requires that $\lambda = \max\{\lambda_2, \lambda_3\}$ converges at the rate of $\sqrt{n}$. A reviewer raised a natural and important question: do the data-driven choices of the smoothing parameters achieve this rate? To investigate, we conducted additional simulations using the examples in Section 5.1 with $\sigma = 0.5$ and $\omega = 0.5$, and in Section 5.2 with $c_1 = 0.5$ and $c_2 = 10$, across varying sample sizes n. The smoothing parameter λ was selected using CV, LOOKL, and BIC. As shown in Figure 4, the estimates of $1/\lambda$ increases approximately linearly with $\sqrt{n}$, empirically supporting the theoretical rate required by Theorem 1.

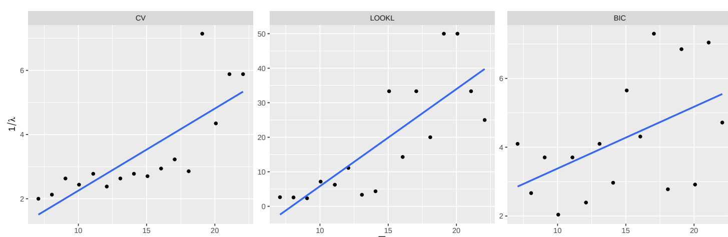

Figure 4: Empirical convergence of data-driven choices of the tuning parameters for density estimation (CV and LOOKL) and edge section (BIC).

6. Application

We apply the cSScGG procedure to study the conditional relationships between some clinical, laboratory, and dialysis variables collected from hemodialysis patients. All patients who underwent dialysis treatments at the Fresenius Medical Care - North America during 2010-2014 are considered. We include patients who stayed at the same facility throughout the treatments. To avoid large fluctuations in the first year on dialysis, we use the average measurements in the second year on dialysis from patients who survived longer than two years. For homogeneity, we include white, non-diabetic, and non-Hispanic patients. After removing missing values, we have $n = 2959$ observations (patients) on the following 27 variables in 3 categories:

Clinical: `age` (years), `height` (cm), `weight` (kg), `bmi` (kg/m^2), `sbp`, `dbp` (systolic and diastolic blood pressure, mmHg), `temp` (Celsius);

Laboratory: `albumin` (g/dL), `ferritin` (ng/mL), `hgb` (hemoglobin, g/dL), `lymphocytes` (%), `neutrophils` (%), `nlr` (neutrophils to lymphocytes ratio, unitless), `sna` (serum sodium, mEq/L), `wbc` (white blood cell, 1000/mc);

Dialysis: `qb` (blood flow, mL/min), `qd` (dialysis flow, mL/min), `saline` (mL), `txttime` (treatment time, min), `olc` (on-line clearance, unitless), `idwg` (interdialytic weight gain, kg), `ufv` (ultrafiltration volume, L), `ufr` (ultrafiltration rate, mL/hr/kg), `epodose` (erythropoietin dose, unit), `volume` (L), `enpcr` (equilibrated normalized protein catabolic rate, g/kg/day), `ektv` (equilibrated Kt/V). Note that `nlr` and `epodose` have been transformed to make them close to Gaussian. In particular, `nlr` is the logarithm of the neutrophils to lymphocytes ratio, and `epodose` is the 1/4 power transformation of the actual erythropoietin dose.

The primary objective of this study is to discover the interactions between all

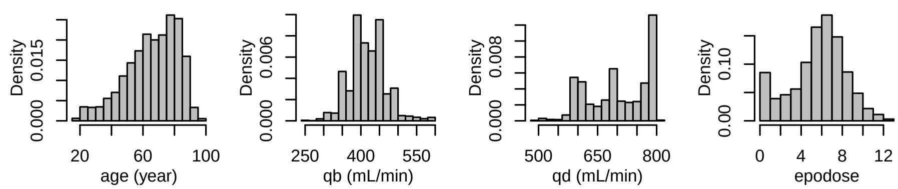

Figure 5: Histograms of `age`, `qb`, `qd` and `epodose`.

these measurements. We first check the marginal distributions of all 27 variables to investigate possible violation of the Gaussian assumption. We identify 4 variables, `age`, `qb`, `qd` and `epodose` as non-Gaussian with very small p-values (less than 2×10^{-16}). Histograms in Figure 5 indicate that the distribution of `age` is skewed, and the distributions of `qb`, `qd` and `epodose` have multiple peaks. Note that despite the 1/4 power transformation, the distribution of `epodose` is still far from normal due to the point mass at zero. Consequently, we specify these 4 variables as $\boldsymbol{X}$ to be estimated nonparametrically.

We compare cSScGG with Glasso and NPN. The estimated graph structures are shown in Figure 6. Visually, there is a large set of edges shared by cSScGG and Glasso, due to the fact that cSScGG assumes the majority of the variables are conditionally normal. However, the graph of Glasso is much denser. To see how cSScGG differs from the other two methods, Figure 7 shows edges detected by the cSScGG procedure only. It shows that the `bmi` is a hub node whose connections with other variables such as `age`, `dbp`, and `wbc` are not selected by other methods. Meanwhile, `qb` has multiple connections with nodes from the other two categories (Clinical and Laboratory). The `bmi` in dialysis patients is intricately linked to a multitude of factors, including age, nutritional status, body composition, inflammation, and dialysis adequacy. It is therefore not surprising that `bmi` functions as a hub node, connected to various clinical variables such as `age`, `dbp`, and `wbc` [21, 22, 23, 24]. In the context of dialysis, `qb` represents the rate at which blood flows through the dialyzer, a parameter crucial for effective treatment. Generally, higher `qb` values improve toxin removal, thereby enhancing dialysis adequacy and efficiency. However, elevated `qb` can adversely affect venous pressures and the health of the vascular access. Consequently, `qb` exhibits multiple connections with various clinical and laboratory variables [24]. These connections identified by the proposed methodology underscore the multifaceted nature of both `bmi` and `qb` as key indicators in dialysis patients. The value of these extra edges remains to be further explored.

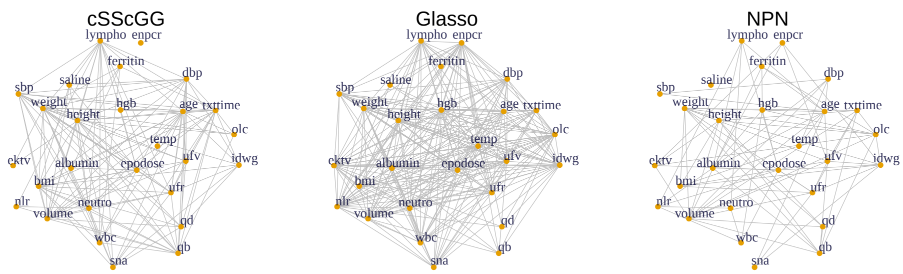

Figure 6: The estimated graphs using cSScGG (*left*), Glasso (*middle*), and NPN (*right*). Tuning parameters are selected by the BIC method. The layout of nodes is fixed across four topologies.

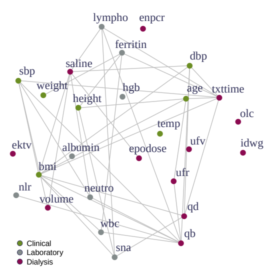

Figure 7: Edges exclusively detected by cSScGG. The measurements are grouped into three categories as described at the beginning of Section 6.

7. Conclusion

This paper develops a new semiparametric density estimation framework called the cSScGG model, which combines the flexibility of smoothing splines and the succinctness of cGGM, allowing us to model high-dimensional data without assuming a joint Gaussian distribution. We established asymptotic convergence properties for both density estimation and edge selection. Simulation results showed that the cSScGG model performs well compared to other existing methods. Applications to real data indicated the cSScGG model could detect edges that may provide new perspectives for researchers.

We used a simple yet effective method to identify non-Gaussian components, which worked reasonably well in our simulations and real data analysis. For future work, we will propose methods that learn the Gaussian variables and graph structure simultaneously. Following the current literature, we have considered continuous $\boldsymbol{X}$ in the construction of the cGGM. We will extend the cGGM and cSScGG to the case where some $\boldsymbol{X}$ components are discrete.

Acknowledgements

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Appendix A. Proofs of Theoretical Results

Proofs of Theorems 2 and 3 are the same as those Luo [9]. We do not repeat here in order to save space.

To prove Theorem 1, we first introduce a sequence of lemmas as in Wytock and Kolter [1]. Note that, different from Wytock and Kolter [1], we allow different penalties for Λ and Θ and a random design matrix X. Lemma 3 below studies the decay rate of the gradients $\nabla_\Theta l_2(\Lambda_0, \Theta_0)$ and $\nabla_\Lambda l_2(\Lambda_0, \Theta_0)$ in element-wise infinity operator norm as sample size increases. Lemmas 1 and 2 enable the extension to a random design matrix and Lemma 4 extends the primal-dual witness approach proposed in Wainwright [15] to our multi-penalties and random design settings.

Lemma 3. *Suppose that Assumption 1 holds. Then*

$$\mathbb{P}(\|\nabla_\Theta l_2(\Lambda_0, \Theta_0)\|_\infty > \vartheta) \leq 2dp \exp\Big\{ -\frac{n\vartheta^2}{8C_\sigma^2 C_s^2}\Big\}, \tag{A.1}$$

$$\mathbb{P}(\|\nabla_\Lambda l_2(\Lambda_0, \Theta_0)\|_\infty > \vartheta) \leq 4p^2 \exp\Big\{ -\frac{n\vartheta^2}{3200C_\sigma^2}\Big\}, \tag{A.2}$$

for any $\vartheta \in (0, 40C_{\Sigma_0})$.

Proof. For the proof of Lemma 3, we modified one step in Wytock and Kolter [1]'s proof by using the maximum diagonal entry of Σ, C_s^2, to bound the column variances of X and also applied the Chernoff bound for the Gaussian tail probability. Ravikumar et al. [25] proved (A.2) in their Lemma 1. □

Proof of Lemma 1.

Proof. The proof is available at `https://github.com/annaliuumass/cSScGG`. □

Proof of Lemma 2. For the jth column of X, $\boldsymbol{X}^j$, $||\boldsymbol{X}^j||_2^2/n = \sum_{i=1}^n X_{ij}^2/n$. Denote $\Sigma_{jj} = E[X_{ij}^2]$. By Chebyshev's inequality,

$$\mathbb{P}(||\boldsymbol{X}^j||_2^2/n > C_X^2) = \mathbb{P}(Z_n > \frac{C_X^2 - \Sigma_{jj}}{\sigma_{4,j}/\sqrt{n}}) \leq \frac{\sigma_{4,j}^2}{n(C_X^2 - \Sigma_{jj})^2},$$

where $Z_n = (||\boldsymbol{X}^j||_2^2/n - \Sigma_{jj})\sqrt{n}/\sigma_{4,j}$. Then, when the sample size $n > \frac{\sigma_{4,j}^2}{\epsilon(C_X^2-\Sigma_{jj})^2}$, we have $\mathbb{P}(||\boldsymbol{X}^j||_2/\sqrt{n} \le C_X) \ge 1-\epsilon$. Let $n_c = \max_{j=1,\ldots,d}\left\{\frac{\sigma_{4,j}^2}{\epsilon(C_X^2-\Sigma_{jj})^2}\right\}$. When $n > n_c$, $||\boldsymbol{X}^j||_2/\sqrt{n} \le C_X$ for all $j = 1, \cdots, d$, with at least probability $1-\epsilon$, which implies that $\mathbb{P}(\max_{j=1,\ldots,d} \left\|\boldsymbol{X}^j\right\|_2 / \sqrt{n} \le C_X) \ge 1-\epsilon$. □

Let $\mathcal{E}$ denote the event $\max_{j=1,\ldots,d} \left\|\boldsymbol{X}^j\right\|_2 / \sqrt{n} \le C_X$, Lemma 2 implies $\mathbb{P}(\mathcal{E}) \ge 1-\epsilon$. For simplicity of notations, we condition on $\mathcal{E}$ for the rest of the proofs.

Let $\Gamma = (\Lambda^T, \Theta^T)^T$. With a bit of abuse of notation, let $l_2(\Gamma) = l_2(\Lambda, \Theta)$.

Lemma 4. *Suppose that the true parameter Γ_0 has support S. We consider two optimization problems:*

$$\hat{\Gamma} = \arg\min_{\Gamma} \Big\{ l_2(\Gamma) + \lambda(\|\Lambda\|_1 + r\,\|\Theta\|_1) \Big\}, \tag{A.3}$$

$$\tilde{\Gamma} = \arg\min_{\Gamma, \Gamma_{\bar{S}=0}} \Big\{ l_2(\Gamma) + \lambda(\|\Lambda\|_1 + r\,\|\Theta\|_1) \Big\}. \tag{A.4}$$

Let $\Delta = \tilde{\Gamma} - \Gamma_0$ and $R(\Delta) = \nabla^2_\Gamma l_2(\Gamma_0)\Delta + \nabla_\Gamma l_2(\Gamma_0) - \nabla_\Gamma l_2(\tilde{\Gamma})$. If the following conditions hold, and the sample size $n > \big(\max\{C_*, C_{**}\}(\log\log d)^2 d^{1/2-2/q}\big)^{-1/(2/q-1/2)}$,

1. *the solution $\tilde{\Gamma}$ is unique;*
2. $\left\|\left\|\left\| (H_{0,\Sigma})_{\bar{S}S} (H_{0,\Sigma})_{SS}^{-1} \right\|\right\|\right\|_\infty < C_0(1-\alpha)$ *for $0 < C_0 \le r \le 1$, $0 < \alpha < 1$;*
3. $\max\{\|\nabla_\Gamma l_2(\Gamma_0)\|_\infty, \|R(\Delta)\|_\infty\} \le \frac{C_0\alpha\lambda}{8}$*;*

then the two ℓ_1-regularized solutions are identical, $\tilde{\Gamma} = \hat{\Gamma}$ with at least probability $1-\epsilon$, where ϵ is a fixed accuracy close to zero.

Proof. Define $\Delta_\Lambda = \tilde{\Lambda} - \Lambda_0$, $\Delta_\Theta = \tilde{\Theta} - \Theta_0$ and $\Delta = (\Delta_\Lambda^T, \Delta_\Theta^T)^T$. Let $R(\Delta) = (R_\Lambda^T(\Delta_\Lambda, \Delta_\Theta),$
$R_\Theta^T(\Delta_\Lambda, \Delta_\Theta))^T$ be the residual of the second-order Taylor expansion of the log-likelihood where

$$\begin{aligned}
& R_\Lambda(\Delta_\Lambda, \Delta_\Theta) \\
= \; & \nabla^2_\Lambda l_2(\Lambda_0, \Theta_0)\Delta_\Lambda + \nabla_\Theta\nabla_\Lambda l_2(\Lambda_0, \Theta_0)\Delta_\Theta + \nabla_\Lambda l_2(\Lambda_0, \Theta_0) \\
& -\nabla_\Lambda l_2(\Lambda_0 + \Delta_\Lambda, \Theta_0 + \Delta_\Theta), R_\Theta(\Delta_\Lambda, \Delta_\Theta) \\
= \; & \nabla^2_\Theta l_2(\Lambda_0, \Theta_0)\Delta_\Theta + \nabla_\Lambda\nabla_\Theta l_2(\Lambda_0, \Theta_0)\Delta_\Lambda + \nabla_\Theta l_2(\Lambda_0, \Theta_0) \\
& -\nabla_\Theta l_2(\Lambda_0 + \Delta_\Lambda, \Theta_0 + \Delta_\Theta).
\end{aligned}$$

Following the same arguments as in Lemma 3 in Ravikumar et al. [25], the ℓ_1 optimization problem (A.3) satisfies

$$\nabla^2_\Gamma l_2(\Gamma_0)\Delta + \nabla_\Gamma l_2(\Gamma_0) - R(\Delta) + \lambda Z = 0, \tag{A.5}$$

where $Z = (Z_\Lambda^T, rZ_\Theta^T)^T$ is the sub-differential of the penalty term evaluated at Λ and Θ, and

$$Z_{\Lambda,ij} = \begin{cases} 0 & \text{if } i = j \\ \mathrm{sign}(\Lambda_{ij}) & \text{if } i \ne j \text{ and } \Lambda_{ij} \ne 0 \\ \in [-1, 1] & \text{if } i \ne j \text{ and } \Lambda_{ij} = 0, \end{cases}$$

$$Z_{\Theta,ij} = \begin{cases} \text{sign}(\Theta_{ij}) & \text{if } \Theta_{ij} \neq 0 \\ \in [-1,1] & \text{if } \Theta_{ij} = 0. \end{cases}$$

If we can verify the strict dual feasibility $\left\|Z_{\bar{S},\Lambda}\right\|_\infty, \left\|Z_{\bar{S},\Theta}\right\|_\infty < 1$, then by Lemma 3 in Ravikumar et al. [25], the restricted solution $\tilde{\Gamma}$ is an optimal solution to the original ℓ_1 problem, i.e., $\tilde{\Gamma} = \hat{\Gamma}$.

Denoting $H = \nabla^2_\Gamma l_2(\Gamma_0)$ and $G = \nabla_\Gamma l_2(\Gamma_0)$ for simplicity, the optimality condition of (A.5) in terms of S and $\bar{S}$ can be rewritten as

$$\begin{bmatrix} H_{SS} & H_{S\bar{S}} \\ H_{\bar{S}S} & H_{\bar{S}\bar{S}} \end{bmatrix} \begin{bmatrix} \Delta_S \\ 0 \end{bmatrix} + \begin{bmatrix} G_S \\ G_{\bar{S}} \end{bmatrix} - \begin{bmatrix} R(\Delta)_S \\ R(\Delta)_{\bar{S}} \end{bmatrix} + \lambda \begin{bmatrix} Z_S \\ Z_{\bar{S}} \end{bmatrix} = 0. \tag{A.6}$$

Since H_{SS} is invertible, we have

$$\Delta_S = H_{SS}^{-1}(R(\Delta)_S - G_S - \lambda Z_S). \tag{A.7}$$

Plugging (A.7) back into the second equation in (A.6), we obtain

$$\begin{bmatrix} Z_{\bar{S},\Lambda} \\ rZ_{\bar{S},\Theta} \end{bmatrix} = -\frac{1}{\lambda} H_{\bar{S}S} H_{SS}^{-1}(R(\Delta)_S - G_S) + H_{\bar{S}S} H_{SS}^{-1} \begin{bmatrix} Z_{S,\Lambda} \\ rZ_{S,\Theta} \end{bmatrix} + \frac{1}{\lambda}(R(\Delta)_{\bar{S}} - G_{\bar{S}}). \tag{A.8}$$

Decompose the first term, $H_{\bar{S}S}$, H_{SS}^{-1}, and the third term corresponding to $Z_{\bar{S},\Lambda}$ and $Z_{\bar{S},\Theta}$, (A.8) can be written as

$$\begin{aligned} Z_{\bar{S},\Lambda} =& \left\{ -\frac{1}{\lambda} H_{\bar{S}S} H_{SS}^{-1}(R(\Delta)_S - G_S) \right\}_\Lambda + H^\Lambda_{\bar{S}S} {H_{SS}^{-1}}^\Lambda Z_{S,\Lambda} + rH^\Lambda_{\bar{S}S} {H_{SS}^{-1}}^\Theta Z_{S,\Theta} \\ &+ \left\{ \frac{1}{\lambda}(R(\Delta)_{\bar{S}} - G_{\bar{S}}) \right\}_\Lambda \\ Z_{\bar{S},\Theta} =& \frac{1}{r} \left\{ -\frac{1}{\lambda} H_{\bar{S}S} H_{SS}^{-1}(R(\Delta)_S - G_S) \right\}_\Theta + \frac{1}{r} H^\Theta_{\bar{S}S} {H_{SS}^{-1}}^\Lambda Z_{S,\Lambda} + H^\Theta_{\bar{S}S} {H_{SS}^{-1}}^\Theta Z_{S,\Theta} \\ &+ \frac{1}{r} \left\{ \frac{1}{\lambda}(R(\Delta)_{\bar{S}} - G_{\bar{S}}) \right\}_\Theta . \end{aligned}$$

When $n > \left(\max\{C_*, C_{**}\}(\log\log d)^2 d^{1/2-2/q}\right)^{-1/(2/q-1/2)}$, apply l_∞ norm and $|||H_{\bar{S}S} H_{SS}^{-1}|||_\infty < |||H_{\Sigma,\bar{S}S} H_{\Sigma,SS}^{-1}|||_\infty + C_\alpha$, then

$$\begin{aligned} ||Z_{\bar{S},\Lambda}||_\infty <& \frac{C_0(1-\alpha) + C_\alpha + 1}{\lambda}(||G||_\infty \\ &+ ||R(\Delta)||_\infty) + || \begin{bmatrix} H^\Lambda_{\bar{S}S} {H_{SS}^{-1}}^\Lambda & H^\Lambda_{\bar{S}S} {H_{SS}^{-1}}^\Theta \end{bmatrix} \begin{bmatrix} Z_{S,\Lambda} \\ Z_{S,\Theta} \end{bmatrix} ||_\infty \\ \le& \frac{C_0(1-\alpha) + C_\alpha + 1}{\lambda} \frac{C_0 \alpha \lambda}{4} + C_0(1-\alpha) + C_\alpha \\ \le& \frac{2-\alpha}{\lambda} \frac{C_0 \alpha \lambda}{4} + (1-\alpha) + \frac{C_\alpha C_0 \alpha}{4} + C_\alpha \\ \le& (2-\alpha)\frac{\alpha}{4} + (1-\alpha) + \frac{C_\alpha \alpha}{4} + C_\alpha / C_0 = 1, \end{aligned} \tag{A.9}$$

$$\begin{aligned}
||Z_{\bar{S},\Theta}||_\infty &\leq \frac{2-\alpha}{r\lambda}(||G||_\infty + ||R(\Delta)||_\infty) + \frac{1}{r}|| \begin{bmatrix} H^{\Theta}_{\bar{S}S}H^{-1}_{SS}{}^{\Lambda} & H^{\Theta}_{\bar{S}S}H^{-1}_{SS}{}^{\Theta} \end{bmatrix} \begin{bmatrix} Z_{S,\Lambda} \\ Z_{S,\Theta} \end{bmatrix} ||_\infty \\
&< \frac{C_0(1-\alpha)+C_\alpha+1}{r\lambda}\frac{r\alpha\lambda}{4} + \frac{C_0(1-\alpha)+C_\alpha}{r} \\
&\leq \frac{2-\alpha}{r\lambda}\frac{r\alpha\lambda}{4} + (1-\alpha) + \frac{C_\alpha\alpha}{4} + C_\alpha/r \\
&\leq (2-\alpha)\frac{\alpha}{4} + (1-\alpha) + \frac{C_\alpha\alpha}{4} + C_\alpha/C_0 = 1. \qquad \text{(A.10)}
\end{aligned}$$

Then $||Z_{\bar{S},\Lambda}||_\infty, ||Z_{\bar{S},\Theta}||_\infty < 1$ hold with probability at least $1-\epsilon$. Note that Lemma 4 applies a generic primal-dual witness approach, which was derived in [15]. By verifying that the strict dual feasibility $||Z_{\bar{S},\Lambda}||_\infty, ||Z_{\bar{S},\Theta}||_\infty < 1$ hold with probability at least $1-\epsilon$, the primal-dual witness construction succeeds, and the solution $\tilde{\Gamma}$ to the restricted problem (A.4) is equal to the solution $\hat{\Gamma}$ to the unrestricted problem (A.3) with probability at least $1-\epsilon$. □

Based on Lemma 4, $\tilde{\Gamma}$ is constructed as a witness to the original unrestricted solution $\hat{\Gamma}$. Then $\tilde{\Gamma}$ inherits many optimality properties from $\hat{\Gamma}$, in terms of the discrepancy to the true Γ_0 and the recovery of the signed sparsity pattern. Our next step is to bound the residual term $\|R(\Delta)\|_\infty$ in terms of $\|\Delta\|_\infty$.

Lemma 5. *(Control of remainder) Suppose that $\mathcal{E}$ is true, $\|\Delta\|_\infty \leq \gamma^{-1}\min\{1/(3C_{\Sigma_0}),$ $C_\Theta/2\}$, then*

$$\|R(\Delta)\|_\infty \leq 206 C^4_{\Sigma_0} C^2_\Theta C^2_X \gamma^2 \|\Delta\|^2_\infty .$$

The proof follows the same steps as in Wytock and Kolter [1].

Lemma 6. *(Control of Δ) Suppose that $u \triangleq 2\kappa_H(\max\{\|\nabla_\Theta l_2(\Lambda_0,\Theta_0))\|_\infty,$ $\|\nabla_\Lambda l_2(\Lambda_0,\Theta_0))\|_\infty\} + \lambda) \leq \min\{1/(3C_{\Sigma_0}\gamma), C_\Theta/(2\gamma), 1/(412\kappa_H C^4_{\Sigma_0} C^2_\Theta C^2_X \gamma^2)\}$. Then*

$$\|\Delta\|_\infty = \left\|\tilde{\Gamma} - \Gamma_0\right\|_\infty \leq u. \qquad \text{(A.11)}$$

Proof. Recall that $\Delta = \tilde{\Gamma} - \Gamma_0$, $\tilde{\Gamma}_{\bar{S}} = \Gamma_{0,\bar{S}} = 0$, therefore $\|\Delta\|_\infty = \|\Delta_S\|_\infty$. Our goal is to bound the deviation Δ. By (A.7), we have $\Delta_S = H^{-1}_{SS}(R(\Delta)_S - G_S - \lambda Z_S)$. In the following, we use Brouwer's fixed-point theorem on a compact set to construct a ball $\mathbb{B}(u)$ that contains Δ. Define the ℓ_∞-ball $\mathbb{B}(u) = \{\Delta | \, \|\Delta_S\|_\infty < u\}$ and a continuous map $\mathcal{F}: \Delta_S \to F(\Delta_S)$ such that

$$F(\Delta_S) = H^{-1}_{SS}(R(\Delta_S) - G_S - \lambda Z_S). \qquad \text{(A.12)}$$

Now it suffices to show $F\big(\mathbb{B}(u)\big) \in \mathbb{B}(u)$, as this implies there is a solution to the above equation. By the uniqueness of the optimal solution, we can thus conclude that Δ belongs in this ball.

Taking infinity norm to (A.12), we have

$$\|F(\Delta_S)\|_\infty \leq \left\|H^{-1}_{SS}\right\|_\infty \|R(\Delta)\|_\infty + \left\|H^{-1}_{SS}\right\|_\infty \|G + \lambda Z\|_\infty . \qquad \text{(A.13)}$$

For any $\Pi \in \mathbb{B}(u)$, by Lemma 5, the first term in (A.13) is bounded by

$$\left\|H^{-1}_{SS}\right\|_\infty \|R(\Pi)\|_\infty \leq \kappa_H 206 C^4_{\Sigma_0} C^2_\Theta C^2_X \gamma^2 \|\Pi\|^2_\infty \leq \frac{u}{2}.$$

By the definition of radius u, the second term in (A.13) is bounded by

$$\left\|H_{SS}^{-1}\right\|_\infty \|G+\lambda Z\|_\infty \le \kappa_H(\|G\|_\infty+\lambda) \le \frac{u}{2}.$$

Therefore, we have $\|F(\Pi)\|_\infty \le u$. $\square$

Proof of Theorem 1 We first show that $\tilde{\Gamma}$ equals $\hat{\Gamma}$ with high probability. Then we proceed with the proof conditioning on this event and the event $\mathcal{E}$.

By Lemma 1, we have the element-wise tail conditions for Λ, $\mathbb{P}(\max_{i,j} |\nabla_{\Lambda,ij} l_2(\Lambda_0, \Theta_0)| > \delta) \le 1/f_\Lambda(n,\delta)$, where $f_\Lambda(n,\delta) = (1/4)\exp\left(n\delta^2/(3200C_\sigma^2)\right)$, and $\nabla_{\Lambda,ij} l_2(\Lambda_0, \Theta_0)$ denotes the (i,j)-th element in $\nabla_\Lambda l_2(\Lambda_0,\Theta_0)$. For a fixed n, denote

$$\bar{\delta}_{f_\Lambda}(n;\omega) = \arg\max_{\delta}\{f_\Lambda(n,\delta) < \omega\}. \tag{A.14}$$

Similarly, for each fixed $\delta > 0$, denote

$$\bar{n}_{f_\Lambda}(\delta;\omega) = \arg\max_{n}\{f_\Lambda(n,\delta) < \omega\}. \tag{A.15}$$

By the monotonicity of the function $f_\Lambda(\delta;n)$, it is easy to see that

$$n > \bar{n}_{f_\Lambda}(\delta;\omega) \text{ for some } \delta > 0 \quad \Longrightarrow \quad \bar{\delta}_{f_\Lambda}(n;\omega) \le \delta. \tag{A.16}$$

Applying Corollary 1 and Lemma 8 in Ravikumar et al. [25], for any $\tau > 2$, we have the control of sampling noise for $\hat{\Lambda}$

$$\mathbb{P}\Big(\|\nabla_\Lambda l_2(\Lambda_0,\Theta_0)\|_\infty > \bar{\delta}_{f_\Lambda}(n;p^\tau)\Big) \le \frac{1}{p^{\tau-2}} \to 0 \tag{A.17}$$

where $\bar{n}_{f_\Lambda}(\delta;p^\tau) = 3200C_\sigma^2(\tau\log p + \log 4)/\delta^2$ and $\bar{\delta}_{f_\Lambda}(n;p^\tau) = \sqrt{3200C_\sigma^2}$ $\sqrt{(\tau\log p+\log 4)/n}$. Now we develop the control of sampling noise for Θ. Again, by Lemma 1, we have the element-wise tail probability for $\hat{\Theta}$:

$$\mathbb{P}\big(\max_{i,j} |\nabla_{\Theta,ij} l_2(\Lambda_0,\Theta_0)| > \delta\big) \le \frac{1}{f_\Theta(n,\delta)}$$

where $f_\Theta(n,\delta) = (1/2)\exp\left(n\delta^2/(8C_\sigma^2C_s^2)\right)$.
Define $\bar{\delta}_{f_\Theta}(n;\omega)$ and $\bar{n}_{f_\Theta}(\delta;\omega)$ similarly to (A.14) and (A.15). Applying the union bound over all pd entries of the gradient matrix, we obtain that

$$\mathbb{P}(\max_{i,j} |\big(\nabla_{\Theta,ij} l_2(\Lambda_0,\Theta_0)| > \delta\big) \le \frac{pd}{f_\Theta(n,\delta)}.$$

Let $\delta = \bar{\delta}_{f_\Lambda}\big(n;(pd)^\tau\big)$, then for any $\tau > 1$,

$$\mathbb{P}\Big(\|\nabla_\Theta l_2(\Lambda_0,\Theta_0)\|_\infty > \bar{\delta}_{f_\Theta}(n;(pd)^\tau)\Big) \le \frac{pd}{f_\Theta\Big(n;\bar{\delta}_{f_\Theta}\big(n;(pd)^\tau\big)\Big)} = \frac{1}{(pd)^{\tau-1}} \to 0, \tag{A.18}$$

which follows from $f_\Theta\Big(n,\bar{\delta}_{f_\Theta}\big(n;(pd)^\tau\big)\Big) = (pd)^\tau$, based on the definition of $\bar{\delta}_{f_\Theta}$.

Calculation shows that $\bar{n}_{f_\Theta}\big(\delta;(pd)^\tau\big)$ and $\bar{\delta}_{f_\Theta}\big(n;(pd)^\tau\big)$ take the forms

$$\bar{n}_{f_\Theta}\big(\delta;(pd)^\tau\big) = 8C_\sigma^2 C_s^2\big(\frac{\tau\log(pd)+\log 2}{\delta^2}\big)$$

and

$$\bar{\delta}_{f_\Theta}\big(n;(pd)^\tau\big) = \sqrt{8C_\sigma^2 C_s^2}\sqrt{\frac{\tau\log(pd)+\log 2}{n}}.$$

Let $\bar{n}_{f_\Gamma} = \max\{\bar{n}_{f_\Lambda}, \bar{n}_{f_\Theta}\}$, $\bar{\delta}_{f_\Gamma} = \max\{\bar{\delta}_{f_\Lambda}, \bar{\delta}_{f_\Theta}\}$. By (A.17-A.18), we have

$$\mathbb{P}\big(\max\{\|\nabla_\Lambda l_2(\Lambda_0,\Theta_0)\|_\infty, \|\nabla_\Theta l_2(\Lambda_0,\Theta_0)\|_\infty\} < \bar{\delta}_{f_\Gamma}\big) \geq 1-\big(p^{-(\tau-2)}+(pd)^{-(\tau-1)}\big). \tag{A.19}$$

Specifically, let $C_s^\star = \max\{C_s, 1\}$,

$$\begin{aligned}\bar{\delta}_{f_\Gamma} &= \max\Big\{\sqrt{3200C_\sigma^2}\sqrt{\frac{\tau\log p+\log 4}{n}}, \sqrt{8C_\sigma^2C_s^2}\sqrt{\frac{\tau\log(pd)+\log 2}{n}}\Big\}\\ &\leq C_\sigma C_s^\star\sqrt{3200}\sqrt{\frac{\tau\log(pd)+\log 4}{n}}.\end{aligned} \tag{A.20}$$

Let $\mathcal{A}$ denote the event $\max\{\|\nabla_\Lambda l_2(\Lambda_0,\Theta_0)\|_\infty, \|\nabla_\Theta l_2(\Lambda_0,\Theta_0)\|_\infty\} < \bar{\delta}_{f_\Gamma}$, (A.19) implies $\mathbb{P}(\mathcal{A}) \geq 1-\big(p^{-(\tau-2)}+(pd)^{-(\tau-1)}\big)$. We condition on the event $\mathcal{A}$ and $\mathcal{E}$ in the following analysis.

Next, we verify that the third assumption in Lemma 4 holds. Choose the (larger) regularization penalty $\lambda = (8/C_0\alpha)\bar{\delta}_{f_\Gamma}$, then the first half $\|\nabla_\Gamma l_2(\Gamma_0)\|_\infty \leq C_0\alpha\lambda/8$ is satisfied. It remains to establish the bound $\|R(\Delta)\|_\infty \leq C_0\alpha\lambda/8$. We do so by using Lemmas 6 and 5 consecutively. Choose

$$\delta = \frac{1}{2\kappa_H}\big(1+\frac{8}{C_0\alpha}\big)^{-2}\min\Big\{\frac{1}{3C_{\Sigma_0}\gamma}, \frac{C_\Theta}{2\gamma}, \frac{1}{412\kappa_H C_{\Sigma_0}^4 C_\Theta^2 C_X^2\gamma^2}\Big\},$$

by our choice of λ, the minimum bound on n and (A.16), we have

$$2\kappa_H\big(1+\frac{8}{C_0\alpha}\big)^2\bar{\delta}_{f_\Gamma} \leq \min\Big\{\frac{1}{3C_{\Sigma_0}\gamma}, \frac{C_\Theta}{2\gamma}, \frac{1}{412\kappa_H C_{\Sigma_0}^4 C_\Theta^2 C_X^2\gamma^2}\Big\}.$$

Applying Lemma 6, we conclude that

$$\|\Delta\|_\infty \leq 2\kappa_H\big(1+\frac{8}{C_0\alpha}\big)\bar{\delta}_{f_\Gamma} \leq 2\kappa_H\big(1+\frac{8}{C_0\alpha}\big)^2\bar{\delta}_{f_\Gamma} \leq \frac{1}{\gamma}\min\Big\{\frac{1}{3C_{\Sigma_0}}, \frac{C_\Theta}{2}\Big\}. \tag{A.21}$$

Then Lemma 5 gives

$$\begin{aligned}\|R(\Delta)\|_\infty &\leq 206C_{\Sigma_0}^4C_\Theta^2C_X^2\gamma^2\|\Delta\|_\infty^2 \leq 824C_{\Sigma_0}^4C_\Theta^2C_X^2\gamma^2\kappa_H^2(1+\frac{8}{C_0\alpha})^2\bar{\delta}_{f_\Gamma}^2\\ &= \Big(824C_{\Sigma_0}^4C_\Theta^2C_X^2\gamma^2\kappa_H^2(1+\frac{8}{C_0\alpha})^2\bar{\delta}_{f_\Gamma}\Big)\frac{C_0\alpha\lambda}{8} \leq \frac{C_0\alpha\lambda}{8}\end{aligned}$$

which follows from the lower bound on n, and the monotonicity (A.16).

To summarize, we have shown that condition 3 in Lemma 4 holds. Furthermore, let S_i denote the nonzero indices of the i-th column of Θ_0, $d < n$ implies that $\lambda_{\min}(1/nX_{S_i}^T X_{S_i}) > 0$, and therefore condition 1 holds. Condition 2 is implied by Assumption 3 and Lemma 1 These allow us to conclude that $\tilde{\Gamma} = \hat{\Gamma}$. By (A.20) and (A.21), the estimator $\hat{\Gamma}$ satisfies the ℓ_∞ bound claimed in Theorem 1(a). Moreover, by the bound (A.11) and the definition of u in Lemma 6, the estimate $\tilde{\Gamma}_{ij}$ cannot differ enough from $\Gamma_{0,ij}$ to change sign when condition (19) is satisfied. This proves Theorem 1(b).

Proof of Corollary 1 Let $\psi = 2\kappa_H(1+\frac{8}{C_0\alpha})C_\sigma C_s^\star\sqrt{3200}\sqrt{\big(\tau\log(pd)+\log 4\big)/n}$. From Theorem 1, we have $\max\left\{\left\|\hat{\Lambda}-\Lambda_0\right\|_\infty, \left\|\hat{\Theta}-\Theta_0\right\|_\infty\right\} \leq \psi$ with probability at least $1-\big(p^{-(\tau-2)}+(pd)^{-(\tau-1)}\big)$. Since Λ_0 has at most $p+s_\Lambda$ non-zeros including diagonal elements and Θ_0 has at most s_Θ non-zeros elements, we have

$$\begin{aligned}
\left|\!\left|\!\left|\hat{\Lambda}-\Lambda_0\right|\!\right|\!\right|_F &= \Big(\sum_{i=1}^{p}(\hat{\Lambda}_{ii}-\Lambda_{0,ii})^2 + \sum_{(i,j)\in E}(\hat{\Lambda}_{ij}-\Lambda_{0,ij})^2\Big)^{1/2} \leq \psi\sqrt{p+s_\Lambda}, \\
\left|\!\left|\!\left|\hat{\Theta}-\Theta_0\right|\!\right|\!\right|_F &= \Big(\sum_{(i,j)\in E}(\hat{\Theta}_{ij}-\Theta_{0,ij})^2\Big)^{1/2} \leq \psi\sqrt{s_\Theta}.
\end{aligned}$$

Combining the above two inequalities leads to the bound in (20).

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